

EN2076 Problems on EM Waves (Sections 7 and 8)

* = confidence builder, ** = exam standard, *** = harder questions, or extracurricular

Note 1: outline numerical solutions are given on the last page of each problem sheet.

Note 2: all numerical answers should be given to 3 s.f., based on the precision of values in questions.

Note 3: four years of past exam questions are provided in a separate document; “EM Waves” appears as part of Q3 on these exam papers (the other part of Q3 being on “Antennas”; I do not issue answers to exam questions but will cover their solution in the revision lectures leading up to the exam.

AP, February 2024

Problems 7 – EM Waves 1

7.1* (See worked example 7.1) A far infrared TEM wave of frequency 5THz propagates in free space along the +x-axis. Calculate the period, phase velocity, wavelength and phase constant of the wave; also sketch the wavefronts (separated by a phase angle π , or 180°).

7.2* (See worked example 7.2) Blue light has a frequency 750THz and wavelength 262nm. Calculate the angular frequency ω , phase constant β and the ratio ω/β ; comment on your result.

7.3* (See worked example 7.3) Microwaves of frequency 5.8GHz propagate in free space, parallel to the +z-axis. Sketch the wavefronts, assuming TEM waves. Show the orientations of $\underline{E}$ and $\underline{H}$ relative to the propagation direction on a single wavefront at times of $t = 0, T/2, T$ and $2T$, clearly labelling how far the wavefront has travelled over these times ($T = 1/f$ is the period of the wave).

7.4/** (See worked example 7.4)** At any instant a TEM wave will propagate along the direction of $\underline{S} = \underline{E} \times \underline{H}$ (called the Poynting vector, S being the power flux, unit W/m^2 ; see Section 8). Use this information to answer the following questions, sketching the orientations of $\underline{E}$, $\underline{H}$ and $\underline{S}$ in each case.

(a)* If $\underline{E}$ is along the x-axis, i.e. along $(1, 0, 0)$, and $\underline{H}$ along the y-axis $(0, 1, 0)$, find the direction of $\underline{S}$.

(b)* If $\underline{E}$ is along the z-axis $(0, 0, 1)$ and $\underline{H}$ along the y-axis $(0, 1, 0)$, find the direction of $\underline{S}$.

(c)* If $\underline{H}$ is along the x-axis $(1, 0, 0)$ and $\underline{S}$ along the -y-axis $(0, -1, 0)$, find the direction of $\underline{E}$.

(d)** If $\underline{E}$ is along $\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$ and $\underline{S}$ along $\left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right)$, find the direction of $\underline{H}$.

7.5/** (See worked example 7.5)** Write down phasor forms of all $\underline{E}$ and $\underline{H}$ components in Problem 7.4(a)-(d), assuming red light in free space in each case (wavelength 680nm) and EM field amplitudes of $E_0 = 524 \text{ V/m}$, $H_0 = 1.39 \text{ A/m}$.

7.6/** (See worked example 7.6)** Points A and B have coordinates $(0, 0, 0)\text{cm}$ and $(4, 0, 0)\text{cm}$, respectively, and sit in free space. A 2 GHz microwave TEM wave propagates along $(1, 0, 0)$. Calculate the phase difference $\Delta\theta$ (a) between wavefronts at A and B measured at the same time, and (b) between a wavefront at A measured when $t = 0\text{ps}$, then measured again at A when $t = 75\text{ps}$. If point B is now moved, its position for wavefronts at A and B to be (c) in phase, (d) 180° out of phase.

7.7* (See worked example 7.7) A green laser beam of frequency 580THz propagates in air. It is then coupled to an optical fibre, which has a refractive index of 1.53. Deduce values of the following in both media: (a) frequency, (b) period, (c) wave velocity, (d) wavelength, and (e) phase constant. Also calculate how long it takes the beam to travel along a 25km length of fibre.

7.8 (See worked example 7.8 and Section 7.4)** The $\underline{E}$ -field of a TEM wave in free space is $\underline{E} = 500 \sin\left(1.2 \times 10^9 t + \beta x\right) \hat{a}_y \text{ V/m}$. Find the phase constant β and period T . Sketch a graph showing

the magnitude of the $\underline{E}$ -field as a function of position x at the fixed times of $t = 0, T/4, T/2$ and $3T/4$ (clearly label the propagation direction).

7.9 (See worked example 7.8)** Deduce as much as you can about each of the following two TEM waves from the field components given:

(a) $\underline{E} = (6 \exp j(6 \times 10^{11} t - 2 \times 10^3 z), 0, 0) \text{ V/m}$, (b) $\underline{H} = (0, 0, 0.53 \exp j(3 \times 10^{15} t - 2 \times 10^7 y)) \text{ A/m}$

7.10 Standard Derivations** - derive the following under “exam conditions”, explaining each step.

(a) Starting with Maxwell’s equations in a lossless dielectric, derive the wave equations for the electric and magnetic fields of a TEM wave.

(b) Derive the formula $v_p = f \lambda = \omega / \beta$ for the phase velocity of a TEM wave.

(c) Show that the TEM plane wave forms $E_x = E_0^{+/-} \exp j(\omega t \mp \beta z)$ are solutions of the 1D wave eqn.

(d) Derive the formula for the TEM wave phase velocity in a dielectric $v_p = c / \sqrt{\epsilon_r} \equiv 1 / \sqrt{\epsilon_r \epsilon_0 \mu_0}$.

7.11* For open discussion and developing a deeper understanding**

(a) Why does a TEM wave’s speed reduce on entering a material? Why doesn’t its frequency change?

(b) What happens to the amplitudes of $\underline{E}$ and $\underline{H}$ when a TEM wave crosses a material boundary?

(c) Why refer to the speed of a TEM wave as “phase velocity”? What other velocity could we have?

(d) In the context of TEM waves, what is meant by (i) a dispersive material, (ii) a non-linear material?

(e) When should the “photon” (i.e. particle) description and “wave” description of light be used?

(f) Sunsets are often red. The overhead sky is blue and clouds white. Explain these observations.

(g) “What has Oliver Heaviside ever done for us?” – you may be surprised if you don’t already know!

Problems 7 – Outline Answers to Numerical Problems

7.1 Period $T = 1/f = 0.2 \text{ ps}$, phase velocity $v_p = c = 3 \times 10^8 \text{ m/s}$, wavelength $\lambda = c/f = 60 \mu\text{m}$, phase constant $\beta = 2\pi/\lambda = 1.05 \times 10^5 \text{ rad/m}$. The wavefronts are parallel planes. Those wavefronts with a phase difference of π are separated spatially by $\lambda/2$.

7.2 Angular frequency $\omega = 2\pi f = 4.71 \times 10^{15} \text{ rad/s}$, phase constant $\beta = 2\pi/\lambda = 2.40 \times 10^7 \text{ rad/m}$, the ratio $\omega/\beta = 1.96 \times 10^8 \text{ m/s}$. This ratio is the phase velocity v_p , i.e. the speed of the wavefronts, and since this is less than the speed of light in a vacuum $c = 3.00 \times 10^8 \text{ m/s}$ we conclude that the wave is travelling in a material. The reduction factor is the refractive index n , where here $n = 1.53$ (i.e. glass).

7.3 The wavelength is $\lambda = 5.17 \text{ cm}$. The wavefronts are parallel planes. The wavefronts at times $t = 0$, $T/2$, T and $2T$ would have travelled distances of 0 , $\lambda/2$, λ and 2λ , respectively.

7.4 (a) $\underline{S}$ is along $(0, 0, 1)$; (b) $\underline{S}$ is along $(-1, 0, 0)$; (c) $\underline{E}$ is along $(0, 0, -1)$; (d) $\underline{S}$ is along $(0, 1, 0)$.

7.5 Angular freq $\omega = 2\pi f = 2\pi c/\lambda_0 = 2.77 \times 10^{15} \text{ rad/s}$, phase const $\beta = 2\pi/\lambda_0 = 9.24 \times 10^6 \text{ rad/m}$

(a) $E_x = 524 \exp j(2.77 \times 10^{15}t - 9.24 \times 10^6 z) \text{ V/m}$, $H_y = 1.39 \exp j(2.77 \times 10^{15}t - 9.24 \times 10^6 z) \text{ A/m}$

(b) $E_z = 524 \exp j(2.77 \times 10^{15}t + 9.24 \times 10^6 x) \text{ V/m}$, $H_y = 1.39 \exp j(2.77 \times 10^{15}t + 9.24 \times 10^6 x) \text{ A/m}$

(c) $E_z = -524 \exp j(2.77 \times 10^{15}t + 9.24 \times 10^6 y) \text{ V/m}$, $H_x = 1.39 \exp j(2.77 \times 10^{15}t + 9.24 \times 10^6 y) \text{ A/m}$

(d) Both $\underline{E}$ and $\underline{H}$ have x and z components, of amplitudes $524/\sqrt{2} = 371 \text{ V/m}$, $1.39/\sqrt{2} = 0.983 \text{ A/m}$

$E_x = 371 \exp j(2.77 \times 10^{15}t - 9.24 \times 10^6 y) \text{ V/m}$, $E_z = 371 \exp j(2.77 \times 10^{15}t - 9.24 \times 10^6 y) \text{ V/m}$

$H_x = 0.983 \exp j(2.77 \times 10^{15}t - 9.24 \times 10^6 y) \text{ A/m}$, $H_z = -0.983 \exp j(2.77 \times 10^{15}t - 9.24 \times 10^6 y) \text{ A/m}$

7.6 The phase of a wavefront is $\theta = \omega t - \beta z$. The phase difference between two wavefronts measured at the same time, but at different positions separated by Δz , is $\Delta\theta = -\beta\Delta z$. The phase difference between two wavefronts measured at the same position, but different times delayed by Δt , is $\Delta\theta = \omega\Delta t$. Phase constant $\beta = \omega/v_p = 2\pi f/c = 41.9 \text{ rad/m}$. (a) Measured at A and B the same time

$\Delta\theta = \theta_B - \theta_A = \beta\Delta z = -1.68 \text{ rad} \equiv -96.0^\circ$; (b) Measured at the same position A at different times

$\Delta\theta = \omega\Delta t = 2\pi f \Delta t = 0.943 \text{ rad} \equiv 54.0^\circ$; (c) for wavefronts to be in phase then $\Delta\theta = \theta_B - \theta_A = \beta\Delta z = 2\pi n$

where n is zero or any positive or negative integer, i.e. $\Delta z = n 2\pi/\beta = n\lambda$, so A and B must be separated by 0 cm , $\pm 4 \text{ cm}$, $\pm 8 \text{ cm}$, $\pm 12 \text{ cm}$, etc; (d) for wavefronts to be 180° out of phase we need $\Delta z = n\pi/\beta = n\lambda/2$ where n is any odd integer (positive or negative), so A and B must be separated by $\pm 2 \text{ cm}$, $\pm 6 \text{ cm}$, $\pm 10 \text{ cm}$, etc.

7.7 In air (i.e. approximately a vacuum, or free space): frequency $f = 580 \text{ THz}$, period $T = 1/f = 1.72 \times 10^{-15} \text{ s}$, wave velocity $c = 3.00 \times 10^8 \text{ m/s}$, wavelength $\lambda_0 = c/f = 517 \text{ nm}$, phase constant $\beta_0 = 2\pi/\lambda_0 = 1.22 \times 10^7 \text{ rad/m}$ (note the use of the subscript “0” for the free-space values of quantities that change in different materials). In the optical fibre (made from glass), the frequency and period remain the same as in air (these never change), but wave velocity and wavelength both decrease by $n = 1.53$ (the refractive index), i.e. wave velocity $v_p = c/n = 1.96 \times 10^8 \text{ m/s}$, wavelength

$\lambda = v_p / f = 338\text{nm}$, i.e. λ_0 / n ; the phase constant increases to $\beta = \omega / v_p = n\beta_0 = 1.86 \times 10^7 \text{ rad/m}$. Since EM waves travel at a constant speed, using “speed = distance/time” we find that the time taken for the laser beam to travel a distance ℓ in the glass is $\tau = \ell / v_p = 128\mu\text{s}$ for a 25km length.

7.8 Since the wave propagates in free space (given), $\beta = \omega / c = 4.00 \text{ m}^{-1}$, $f = \omega / 2\pi = 191\text{MHz}$ (i.e. RF signal), $T = 5.24\text{ns}$, wave propagates along $-\hat{a}_x$ with $\underline{E}$ (of amplitude 500 V/m) along $\hat{a}_y$ and $\underline{H}$ along $-\hat{a}_z$. The sketch of the $\underline{E}$ field at different times should be similar to those in Section 7.4.

7.9 (a) $\omega = 6 \times 10^{11} \text{ rad/s}$, $f = \omega / 2\pi = 95.5\text{GHz}$ (i.e. mm wave), $T = 10.5\text{ps}$, $\beta = 2 \times 10^3 \text{ rad/m}$, $\lambda = 2\pi / \beta = 3.14\text{mm}$, $v_p = \omega / \beta = 3 \times 10^8 \text{ m/s}$, wave propagates in free space along $\hat{a}_z$ with $\underline{E}$ along $\hat{a}_x$ (amplitude 6 V/m) and $\underline{H}$ along $\hat{a}_y$ (we’ll calculate its amplitude in Section 8).

(b) $\omega = 3 \times 10^{15} \text{ rad/s}$, $f = 477\text{THz}$, $T = 2.09 \times 10^{-15} \text{ s}$, $\beta = 2 \times 10^7 \text{ rad/m}$, $\lambda = 314\text{nm}$ (i.e. equivalent free space wavelength 628 nm, red light), $v_p = 1.5 \times 10^8 \text{ m/s}$, wave propagates in dielectric of refractive index $n = 2$ along $+\hat{a}_y$ with $\underline{H}$ (of amplitude 0.53 A/m) along $\hat{a}_z$ and $\underline{E}$ along $-\hat{a}_x$. Simple sketches of the relative orientations of $\underline{E}$, $\underline{H}$ and $\underline{S} = \underline{E} \times \underline{H}$ like those of worked example 7.4 will be sufficient.

Problems 8 – EM Waves 2

8.1* (See worked examples 8.1, 8.3) Calculate the wave impedances, all wave amplitudes and rms power fluxes of the two TEM waves of Problem 7.9.

8.2** (See worked example 8.2) The $\underline{H}$ -field of a TEM wave is described by the phasor

$$\underline{H} = (H_x, 0, 0), \quad H_x = 1.45 \exp j(3.46 \times 10^{11} t + 1.82 \times 10^3 y) \text{ A/m}$$

(a) Deduce a similar phasor expression for its $\underline{E}$ -field and extract as much information as you can about with wave from its phasor form. (b) Verify your result for $\underline{E}$ by direct substitution of $\underline{H}$ into M4 (the 4th Maxwell equation).

8.3* (See worked example 8.3) (a) Blue light of wavelength 440nm propagates without reflection in free space with a time averaged power flux of $50.0 \mu\text{W}/\text{cm}^2$. Find the electric and magnetic field amplitudes of the wave. (b) 200 MHz radio waves propagate in PTFE, of relative permittivity 2.04. If the amplitude of the wave's $\underline{H}$ -field is 5.62 A/m, calculate the amplitude of the $\underline{E}$ -field and the peak and rms power fluxes.

8.4** (See worked example 8.5) Microwaves of frequency 1.80 GHz and rms power flux $20.0 \text{ dBm}/\text{cm}^2$ are incident on a flat metal sheet in free space. Find (a) the $\underline{E}$ -field and $\underline{H}$ -field amplitudes of the incident wave, (b) the peak field amplitudes outside the sheet, and (c) the peak-peak distance outside of the metal. (d) Also draw a fully-labelled sketch of the amplitude envelopes of both $\underline{E}$ and $\underline{H}$ fields outside of the sheet. (Note that $0 \text{ dBm} = 1 \text{ mW}$, $10 \text{ dBm} = 10 \text{ mW}$, etc.)

8.5** (See worked example 8.5) Radio waves of frequency 800 kHz are incident normally to a building and a detector is placed 500 m away from the building, in the path of the radio waves. (a) Calculate the phase difference between the $\underline{E}$ fields of the incident and reflected waves at the detector, measured at the same instant in time, assuming that 180° phase change occurs on reflection. (b) If the detector is moved towards the building along the direction of wave propagation, how far should it be moved for the incident and reflected waves to be exactly in phase, and exactly 180° out of phase. (c) Relate this to the problem of mobile communications within an urban environment.

8.6** (See worked example 8.6) (a) Calculate the time-averaged transmitted and reflected power fluxes when the TEM wave of problem 8.3(a) is normally incident on a glass block, of refractive index 1.52. (b) Calculate the time-averaged transmitted and reflected power fluxes when the TEM wave of problem 8.3(b) is normally incident on a boundary with free space.

8.7** (See Section 8.8) Sketch fully labelled graphs of the $\underline{E}$ and $\underline{H}$ field envelopes in all regions for the TEM wave reflections of problems 8.6(a) and 8.6(b).

8.8** (See worked example 8.7) A 5THz TEM wave of rms power flux $-13.7 \text{ dBm}/\text{cm}^2$ is incident normally from air into a flat alumina substrate, of relative permittivity 10.5. Calculate:

- (a) The EM field amplitudes of incident, reflected and transmitted waves.
- (b) The rms power fluxes of the reflected and transmitted waves.
- (c) Sketch the $\underline{E}$ -field and $\underline{H}$ -field envelopes in the partial standing wave formed in the air region.

8.9 Standard Derivations** - derive the following under “exam conditions”, explaining each step. (a) Show that the wave impedance of a TEM wave is $\eta = E_x / H_y = \eta_0 / n$ (make sure you can do this using either M3, as in the notes, or M4).

(b) Use M3 or M4 to derive that for the forward (incident) TEM wave $E_0^+ = \eta H_0^+$, but for the backward (reflected) TEM wave $E_0^- = -\eta H_0^-$.

(c) Show that for TEM wave propagation in a lossless dielectric $\eta = \sqrt{\mu_0 / \epsilon_r \epsilon_0} \approx 377 / n \, \Omega$.

(d) Derive the formula for attenuation depth in a low-loss dielectric $D \approx \lambda_0 / \pi \sqrt{\epsilon_1} \tan \delta$

(e) Derive formulae for the amplitude envelopes of the electric and magnetic fields of the standing wave outside a flat metal sheet when a TEM is incident normally to the sheet.

(f) Derive the formulae for amplitude reflection and transmission coefficients at a dielectric interface.

8.10* (non-exam)** Show that for a TEM wave with $\underline{E} = (E_x, 0, 0)$ and $\underline{H} = (0, H_y, 0)$, the time-averaged electric and magnetic energy densities are equal (hint: use the fact that $\eta = E_x / H_y = \sqrt{\mu_0 / \epsilon_r \epsilon_0}$). Hence, by calculating the total energy through an area of 1 m^2 in 1 s , show that the power flux of a TEM wave has magnitude $S_z = E_x H_y$. (This is an example of Poynting’s theorem in action for a dielectric in which there is negligible power dissipation; the only way for there to be a flux of power in a particular direction is from the reduction in EM stored field energy).

8.11/** (See worked example 8.8) – non exam** A commercial dielectric used in high frequency printed circuit boards has a complex permittivity of $\epsilon_r = 2.3 - j0.0085$ at 50 GHz. (a) Calculate its loss tangent and estimate its attenuation depth at 50 GHz. (b) Estimate its power attenuation coefficient at 50 GHz, in units of dB/m. (c) Write down phasor forms of the $\underline{E}$ and $\underline{H}$ field components of a 50 GHz TEM wave propagating in this dielectric for a power flux of 1 mW/m^2 , along the +z-axis.

8.12* For open discussion and developing a deeper understanding**

- Explain why broken glass is less easy to see when lying on the floor of a water-filled swimming pool compared to when it lies on the ground next to the pool.
- Explain why sea water is highly absorbing of radio waves and microwaves (absorption depth of a few cm) but very transparent for visible light (absorption depth of $> 100 \text{ m}$).
- The optimal frequency for absorption of microwaves by water at room temperature is 20 GHz but domestic microwave ovens work at 2.45 GHz, industrial ovens at 900 MHz. Why is this?
- What IR wavelengths are used in optical fibre communications; what determines this choice?
- Metals are highly reflective of all EM waves until frequencies reach the UV part of the spectrum and above, when their reflectivity dramatically reduces. What causes this phenomenon?
- Extend the discussion of (e) above to explain how transparent conductors work, used in optoelectronic displays (the paradox to resolve here is that a material is needed that is both transparent and electrically conductive – good metals are highly reflecting to visible light!)
- What design/material features give stealth bombers a very small radar cross section?
- Explain how rainbows are formed, also accounting for multiple rainbows.
- Why are glass lenses coated (i.e. “bloomed”) to reduce their reflection and so increase their light transmission? What is the link between the thickness of the coating and the wavelength, and what refractive index should be used for the coating?
- What is a “Salisbury screen” and how does it work?

Problems 8 – Outline Answers to Numerical Questions

8.1 (a) $E_0 = 6 \text{ V/m}$, $H_0 = E_0 / \eta_0 = 1.59 \times 10^{-2} \text{ A/m}$, $S_{\text{rms}} = \frac{1}{2} E_0 H_0 = 0.0477 \text{ W/m}^2$

(b) $H_0 = 0.53 \text{ A/m}$, $E_0 = \eta H_0 = 99.9 \text{ V/m}$, $S_{\text{rms}} = \frac{1}{2} E_0 H_0 = 26.5 \text{ W/m}^2$

8.2 (a) The wave is propagating along $(0, -1, 0)$ and at $t = 0$, $z = 0$ its $\underline{H}$ field is along $(1, 0, 0)$ so $\underline{E}$ at this instant $\underline{E}$ is along $(0, 0, -1)$ (since $\underline{E} \times \underline{H}$ has to be along the propagation direction) so we write $\underline{E} = (0, 0, E_z)$. The phase velocity is $v_p = \omega/\beta = 1.90 \times 10^8 \text{ m/s}$, so the refractive index is $n = c/v_p = 1.58$, the wave impedance is $\eta = \eta_0/n = 239\Omega$ and the $\underline{E}$ -field amplitude is $E_0 = \eta H_0 = 346 \text{ V/m}$. The $\underline{E}$ -field as a row vector is therefore

$$\underline{E} = (0, 0, E_z), \quad E_z = -345 \exp j(3.46 \times 10^{11} t + 1.82 \times 10^3 y) \text{ V/m}$$

(b) To derive $\underline{E}$ using Maxwell's equations, we substitute the expression for $\underline{H}$ into M4 and solve:

$$\underline{\nabla} \times \underline{H} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & 0 & 0 \end{vmatrix} = \frac{\partial H_x}{\partial z} \hat{a}_y - \frac{\partial H_x}{\partial y} \hat{a}_z = \epsilon_r \epsilon_0 \frac{\partial \underline{E}}{\partial t} = j\omega \epsilon_r \epsilon_0 \underline{E}$$

Since H_x depends only on t and y (not on z) we can simplify this to

$$\begin{aligned} \underline{E} &= -\frac{1}{j\omega \epsilon_r \epsilon_0} \frac{\partial H_x}{\partial y} \hat{a}_z = -\frac{1}{j\omega \epsilon_r \epsilon_0} \frac{\partial}{\partial y} (H_0 \exp j(\omega t + \beta y)) \hat{a}_z \\ &= -\frac{\beta}{\omega \epsilon_r \epsilon_0} H_0 \exp j(\omega t + \beta y) \hat{a}_z = -\eta H_0 \exp j(\omega t + \beta y) \hat{a}_z \end{aligned}$$

Since $H_0 = 1.45 \text{ A/m}$ and (from (a)) $\eta = 239\Omega$, the final result is

$$\underline{E} = (0, 0, E_z), \quad E_z = -345 \exp j(3.46 \times 10^{11} t + 1.82 \times 10^3 y) \text{ V/m}$$

8.3 (a) SI units $S_{\text{rms}} = 0.5 \text{ W/m}^2 \rightarrow E_0 = \sqrt{2\eta_0 S_{\text{rms}}} = 19.4 \text{ V/m}$, $H_0 = \sqrt{2S_{\text{rms}}/\eta_0} = 5.15 \times 10^{-2} \text{ A/m}$.

(b) Wave impedance of PTFE is $\eta = \eta_0 / \sqrt{\epsilon_r} = 264\Omega$, $E_0 = \eta H_0 = 1.48 \text{ kV/m}$, peak power flux $S_{\text{max}} = E_0 H_0 = 8.34 \text{ kW/m}^2$, rms power flux $S_{\text{rms}} = S_{\text{max}} / 2 = 4.17 \text{ kW/m}^2$.

8.4 (a) In SI units $S_{\text{rms}} = 0.1 \text{ W/cm}^2 = 1000 \text{ W/m}^2$, so the field amplitudes of the incident wave are $E_0^+ = \sqrt{2\eta_0 S_{\text{rms}}} = 868 \text{ V/m}$, $H_0^+ = \sqrt{2S_{\text{rms}}/\eta_0} = 2.30 \text{ A/m}$ (b) The peak fields are double the incident field amplitudes, since a perfect standing wave is set up, i.e. $E_{\text{max}} = 1.74 \text{ kV/m}$, $H_{\text{max}} = 4.60 \text{ A/m}$ (c) the peak-peak separation in the standing wave is $\Delta z = \lambda/2 = v_p/2f = 8.33 \text{ cm}$, which is also the separation between nodes; (d) see the main lecture notes for the standing wave sketches, also the video clips **wave4.avi** and **wave5.avi**.

8.5 (a) $\beta = \omega / v_p = 2\pi f / c = 0.0168 \text{ m}^{-1}$ and $\Delta\phi = 2\beta\ell + \pi = 19.9 \text{ rads} \equiv 1.047 \text{ rads}$ or 60° ; (b) for in phase signals, move detector a distance x towards the building, where x satisfies $2\beta(\ell - x) + \pi = n\pi$ (where n is an even integer), i.e. $x = 31.4 \text{ m}$, 218.8 m , 406.3 m ; for out of phase signals, x satisfies $2\beta(\ell - x) + \pi = n\pi$ (n is an odd integer), i.e. $x = 144.4 \text{ m}$, 322.2 m .

8.6 (a) The amplitude reflection coefficient is $\rho = (1 - n) / (1 + n) = -0.206$ and the amplitude transmission coefficient is $\tau = 1 + \rho = 0.794$; the rms reflected power flux is $S_{\text{ref}} = |\rho|^2 S_{\text{in}} = 0.0213 \text{ W/m}^2$. $S_{\text{ref}} = |\rho|^2 S_{\text{in}} = 42.4 \text{ W/m}^2$, so the rms transmitted power flux is $S_{\text{trans}} = S_{\text{in}} - S_{\text{ref}} = 0.488 \text{ W/m}^2$; (b) The refractive index is $n = \sqrt{\epsilon_r} = 1.428$, so the amplitude reflection coefficient is $\rho = (n - 1) / (n + 1) = +0.176$ (note the order and sign, since here the wave is passing from a high permittivity region to a low one) and the amplitude transmission coefficient is $\tau = 1 + \rho = 1.18$; note that although $\tau = E_2^+ / E_1^+ > 1$, so that the transmitted electric field amplitude E_2^+ is greater than the incident electric field amplitude E_1^+ , because the wave impedance in the PTFE is less than that of free space, the incident power flux is greater than the transmitted power flux, so there is no (as if by magic!) amplification occurring here; the rms reflected power flux is $S_{\text{ref}} = |\rho|^2 S_{\text{in}} = 129 \text{ W/m}^2$, the rms transmitted power flux is $S_{\text{trans}} = S_{\text{in}} - S_{\text{ref}} = 4.04 \text{ kW/m}^2$

8.7 This is a re-labeling of the plot in Section 8.9, based on the information calculated above.

8.8 (a) $S_{\text{inc}} = -13.7 \text{ dBm/cm}^2 = 10^{-1.37} \text{ mW/cm}^2 = 10^{-0.37} \text{ W/m}^2 = 0.427 \text{ W/m}^2$.

In air, $E_0^+ = \sqrt{2S_{\text{inc}}\eta_0} = 17.9 \text{ V/m}$, $H_0^+ = \sqrt{2S_{\text{inc}}/\eta_0} = 0.0476 \text{ A/m}$. The wave impedance in the alumina is $\eta = \eta_0 / \sqrt{\epsilon_r} = 116 \Omega$, so the $\underline{E}$ -field amplitude reflection coefficient is

$$\rho = \frac{\eta - \eta_0}{\eta + \eta_0} = \frac{116 - 377}{116 + 377} = -0.528, \text{ so the reflected wave has}$$

$$E_0^- = \rho E_0^+ = -9.45 \text{ V/m}, H_0^- = -E_0^- / \eta_0 = 0.0251 \text{ A/m}$$

The transmitted wave has $E_1^+ = (1 + \rho)E_0^+ = 8.45 \text{ V/m}$, $H_1^+ = E_1^+ / \eta = 0.0727 \text{ A/m}$

(b) Reflected power $S_{\text{ref}} = -\frac{1}{2} E_0^- H_0^- = 0.119 \text{ W/m}^2$; transmitted power $S_{\text{trans}} = \frac{1}{2} E_1^+ H_1^+ = 0.307 \text{ W/m}^2$

Note that $S_{\text{inc}} = S_{\text{ref}} + S_{\text{trans}}$, as is required by energy (i.e. power) conservation.

(c) See section 8.9 for the EM field envelope plots for the partial standing wave.

8.11 (a) Loss tangent $\tan \delta = 0.0037$, free space wavelength $\lambda_0 = c / f = 6 \text{ mm}$, so attenuation depth

$D = 0.340 \text{ m}$; (b) amplitude attenuation $\propto e^{-z/D}$ so power attenuation $\propto e^{-2z/D}$ and to express this in dB/m we take $10 \log_{10}(e^{-2z/D}) = -8.69z / D = -25.5 \text{ dB/m}$; (c) given $S_{\text{rms}} = 0.001 \text{ W/m}^2$ and the wave impedance $\eta = \eta_0 / \sqrt{\epsilon_r} = 248 \Omega$ the field amplitudes are $E_0 = \sqrt{2\eta S_{\text{rms}}} = 0.704 \text{ V/m}$ and

$H_0 = \sqrt{2S_{\text{rms}}/\eta} = 2.84 \times 10^{-3} \text{ A/m}$, phase constant $\beta = 2\pi\sqrt{\epsilon_r} / \lambda_0 = 1.59 \times 10^3 \text{ m}^{-1}$, so the phasor forms for E_x and H_y are $E_x = 0.704 e^{-2.94z} \exp j(3.14 \times 10^{11} t - 1.59 \times 10^3 z) \text{ V/m}$,

$H_y = E_x / \eta = 2.84 \times 10^{-3} e^{-2.94z} \exp j(3.14 \times 10^{11} t - 1.59 \times 10^3 z) \text{ A/m}$ (note the amplitude decay terms)

Problems to do for homework beforehand

Ex class (DC)	Week 5	5.1, 5.4, 5.6, 5.10
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Problems 5 – Transmission line parameters (R , C and L)

5.1* The resistivity of silver is $1.59 \mu\Omega\text{cm}$ (i.e. $1.59 \times 10^{-8} \Omega\text{m}$), of hard drawn copper is $1.77 \mu\Omega\text{cm}$, gold is $2.44 \mu\Omega\text{cm}$, aluminium is $2.83 \mu\Omega\text{cm}$ and steel wire is $11.8 \mu\Omega\text{cm}$. Calculate the resistance per unit length (i.e. per m) of the following at low frequencies (e.g. 50 Hz):

- (a) copper wire of diameter 250 μm ,
- (b) aluminum wire of diameter 3 mm,
- (c) thin gold track, of width 20 μm and thickness 1 μm ,
- ** (d) aluminium wire of outer diameter 3 mm, with a solid steel core of diameter 1 mm,
- ** (e) copper wire of diameter 250 μm with a silver coating of thickness 1 μm .

5.2** Use the concept of skin depth to explain fully why silver-coated copper or silver-coated steel wires are used in high frequency transmission lines (e.g. those operating at frequencies above 1 GHz). Assuming the formula for the skin depth $\delta = \sqrt{\rho/(\pi\mu_0 f)}$, estimate the minimum thickness for the silver coating for a wire to behave as if it were solid silver at 10 GHz. Also use the concept of skin depth to sketch the resistance per unit length of a solid silver wire, of 1 mm diameter, as a function of frequency, from dc up to microwave frequencies (e.g. 10 GHz). For a high frequency coaxial cable, why is it that only the inner conductor is silver plated?

5.3** Calculate the resistance per unit length (at 50 Hz) of

- (a) an all-aluminium conductor (AAC), consisting of 37 strands, each of diameter 1 mm,
- (b) a steel-reinforced aluminum conductor (ACSR), consisting of 7 central steel strands each of diameter 1 mm, surrounded by 30 aluminium strands each of diameter 1 mm,
- (c) a twin wire line, with identical aluminium wires conductors of radius 2.5 mm with wire axes separated by 80 mm,
- (d) a coaxial line made of copper conductors, with $a = 0.5$ mm, $b = 1.5$ mm, with outer conductor of thickness $t = 0.3$ mm.

5.4** Calculate the capacitance per unit length of the coaxial line of problem 5.3(d) if it is:

- (a) air-spaced,
- (b) filled with a Teflon dielectric spacer, of relative permittivity 2.06.
- (c) Calculate the total capacitance of a 150 m length of Teflon-filled line.
- (d)** If the dielectric strength of Teflon is 90 MV/m, calculate the maximum free line charge per unit length of the Teflon-filled line and hence the maximum working voltage of the line (allow a safety factor of 2).
- (e)*** If the air-space between the conductors of a coaxial line could be evacuated, what now limits the maximum line voltage?

5.5** Calculate the total capacitance of a 45 m long section of the air-spaced twin line of problem 5.3(c). If the dielectric strength of dry air is 3.0 MV/m, estimate is the maximum working voltage of the line (assume a safety factor of 2 and state any approximations that you make). If the relative humidity (i.e. water content) of the air increases, what happens to the line's maximum working voltage?

5.6** Calculate the external inductance (i.e. the inductance due to the magnetic flux density outside of the centre conductor) per unit length of the coaxial line of problem 5.3(d). Does this result depend of the presence of dielectric between the conductors? What is the total external inductance of a 35 m long section of this line?

5.7** Calculate the internal inductance per unit length of the centre conductor of the coaxial line of problem 5.3(d). Why is the internal inductance of the outer conductor often neglected (hint: the outer conductor is usually a thin tube). Hence, calculate the difference in total inductance of a 35 m long section of this coaxial line between low frequency (e.g. 50 Hz) and high frequency (e.g. 10 GHz).

5.8** Calculate the inductance per unit length of the air-spaced twin line of problem 5.3(c) at high frequencies and at low frequencies.

5.9* Design a coaxial line of 75Ω characteristic impedance using a Teflon spacer (of relative permittivity 2.06) and an outer conductor with $b = 50$ mm.

5.10* Calculate the characteristic impedance of the air-spaced twin line of problem 5.3(c). How is the characteristic impedance increased for a fixed wire radius? Repeat the calculation if the wires are embedded in a polymer dielectric, of relative permittivity 2.65.

5.11** The propagation velocity along an air-spaced line is 3×10^8 m/s. Calculate C and L if the line's characteristic impedance is 200Ω . A different 50Ω line is found to have $C = 104$ pF/m. Calculate L and the line's signal propagation velocity. Why is the velocity less than 3×10^8 m/s?

5.12*** An air-spaced transmission line consists of a long metal wire of radius 1 mm, whose axis is parallel to and a distance of 6 mm above an earthed metal sheet. Calculate C and L for this line and hence its characteristic impedance. (Hint: consider the method of images, introduced in problem 2.9).

5.13** Sketch the EM fields on the cross sections of the following lines when they carry an electric signal (a) parallel plate line, (b) coaxial line, and (c) twin wire line. Also indicate the positions where dielectric breakdown is most likely to occur for large line voltages.

5.14** The following derivations could crop up in the EN2076 examination and are covered in the lecture notes. Do each as an exercise under examination conditions.

(a) Derive the capacitance per unit length of a coaxial line $C = \frac{2\pi\epsilon_r\epsilon_0}{\ln(b/a)}$

(b) Derive the capacitance per unit length of a twin wire line $C \approx \frac{\pi\epsilon_r\epsilon_0}{\ln(D/a)}$

(c) Derive the external inductance per unit length of a coaxial line $L = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right)$

(d) Derive the external inductance per unit length of a twin wire line $L \approx \frac{\mu_0}{\pi} \ln \frac{D}{a}$

(e) Derive the internal inductance per unit length of a wire $L_{\text{int}} = \frac{\mu_0}{8\pi}$..

Numerical answers to Problems 5

- 5.1 (a) $0.361\ \Omega/\text{m}$; (b) $4.00\ \text{m}\Omega/\text{m}$; (c) $1.22\ \text{k}\Omega/\text{m}$; (d) $4.37\ \text{m}\Omega/\text{m}$; (e) $0.355\ \Omega/\text{m}$
5.2 Silver thickness of around 1.5 to $2.0\ \mu\text{m}$.
5.3 (a) $0.974\ \text{m}\Omega/\text{m}$; (b) $1.14\ \text{m}\Omega/\text{m}$; (c) $2.88\ \text{m}\Omega/\text{m}$; (d) $28.2\ \text{m}\Omega/\text{m}$
5.4 (a) $50.6\ \text{pF}/\text{m}$; (b) $104\ \text{pF}/\text{m}$; (c) $15.6\ \text{nF}$; (d) $24.7\ \text{kV}$
5.5 $8.03\ \text{pF}/\text{m}$; $361\ \text{pF}$; $26.0\ \text{kV}$
5.6 $220\ \text{nH}/\text{m}$; $7.69\ \mu\text{H}$
5.7 $50\ \text{nH}/\text{m}$ for any solid (non-magnetic) cylindrical wire; $7.70\ \mu\text{H}$ at VHF, $9.45\ \mu\text{H}$ at VLF, difference is $1.75\ \mu\text{H}$
5.8 $1.39\ \mu\text{H}/\text{m}$ at VHF, $1.49\ \mu\text{H}/\text{m}$ at VLF, difference is $100\ \text{nH}/\text{m}$
5.10 $416\ \Omega$; by increasing line separation; $255\ \Omega$
5.11 $667\ \text{nH}/\text{m}$; $16.7\ \text{pF}/\text{m}$; $260\ \text{nH}/\text{m}$; $1.92 \times 10^8\ \text{m/s}$
5.12 $22.4\ \text{pF}/\text{m}$; $496\ \text{nH}/\text{m}$; $47.1\ \Omega$

Problems 4 – Biot-Savart law for magnetic flux density $\underline{B}$

4.1* (A review of Ampère's law from EN1216) Use AL to find the magnitude of $\underline{H}$ and $\underline{B}$ for the following (sketch the field lines for each):

(a) Outside of a long wire of radius a carrying a current I , distance r from the axis.

$$\oint_L \underline{H} \bullet d\underline{\ell} = H \cdot 2\pi r = I \quad \therefore H = \frac{I}{2\pi r} \quad \text{and} \quad B = \mu_0 H = \frac{\mu_0 I}{2\pi r}$$

(b) As for (a), except when $r < a$ not all the current is enclosed by the loop:

$$\oint_L \underline{H} \bullet d\underline{\ell} = H \cdot 2\pi r = I \frac{r^2}{a^2} \quad \therefore H = \frac{Ir}{2\pi a^2} \quad \text{and} \quad B = \frac{\mu_0 Ir}{2\pi a^2}$$

(c) Deep inside a long solenoid of N turns, current I and length ℓ .

$$\oint_L \underline{H} \bullet d\underline{\ell} \approx H \cdot \ell = NI \quad \therefore H \approx \frac{NI}{\ell} \quad \text{and} \quad B \approx \frac{\mu_0 NI}{\ell}$$

(d) Between a pair of current sheets, of current I , width W and separation $d \ll W$.

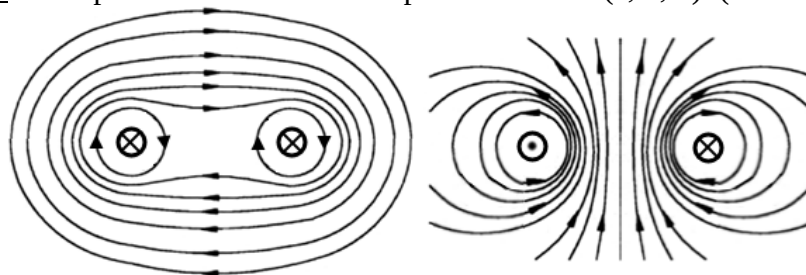
$$\oint_L \underline{H} \bullet d\underline{\ell} \approx H \cdot W = I \quad \therefore H \approx \frac{I}{W} \quad \text{and} \quad B \approx \frac{\mu_0 I}{W}$$

4.2* Find B_z (a) at the centre of a 100 turn circular coil of radius 1 cm carrying a current of 1 A, and (b) deep inside a long solenoid of 2000 turns per metre carrying a current of 0.5 A.

$$(a) B_z = \frac{\mu_0 NI}{2R} = 6.28 \text{ mT}$$

$$(b) B_z = \frac{\mu_0 NI}{\ell} = 1.26 \text{ mT} \quad (N/\ell = 2000)$$

4.3** A long thin wire is aligned parallel to the z -axis with its axis at position vector $(0, 0, 0)$. A second wire is aligned parallel with the first, with its axis at position vector $(d, 0, 0)$. Sketch the lines of $\underline{B}$ around the wires if they carry (a) equal and parallel currents, (b) equal and opposite currents. In each of these two cases, for current magnitudes I , derive a formula for the magnitude of $\underline{B}$ at the point on the x -axis with position vector $(x, 0, 0)$ (assume $0 < x < d$).



For $I_1 = I_2$:

$$B(x) = B_1 - B_2 = \frac{\mu_0 I}{2\pi x} - \frac{\mu_0 I}{2\pi(d-x)} = \frac{\mu_0 I}{2\pi} \left(\frac{1}{x} - \frac{1}{d-x} \right)$$

at $x=d/2$, $B(x)=0$

For $I_1 = -I_2$:

$$B(x) = B_1 + B_2 = \frac{\mu_0 I}{2\pi x} + \frac{\mu_0 I}{2\pi(d-x)} = \frac{\mu_0 I}{2\pi} \left(\frac{1}{x} + \frac{1}{d-x} \right)$$

at $x=d/2$, $B(x) = \frac{2\mu_0 I}{\pi d}$

4.4** Repeat problem 4.3, this time deriving formulae for the magnitude of $\underline{B}$ a long distance from the wires (i.e. at the point with position vector $(x, 0, 0)$, where $|x| \gg d$).

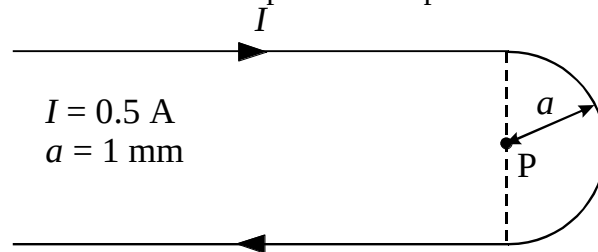
For $I_1 = I_2$, and $x \gg d$

$$B(x) = B_1 - B_2 = \frac{\mu_0 I}{2\pi x} - \frac{\mu_0 I}{2\pi(d-x)} \approx \frac{\mu_0 I}{2\pi} \left(\frac{2}{x} \right) = \frac{\mu_0 I}{\pi x}$$

For $I_1 = -I_2$, and $x \gg d$

$$B(x) = B_1 + B_2 = \frac{\mu_0 I}{2\pi x} + \frac{\mu_0 I}{2\pi(d-x)} \approx \frac{\mu_0 I}{2\pi} \left(\frac{1}{x} - \frac{1}{x} \right) = 0$$

4.5** A long wire is bent into the shape shown in the diagram. Find $\underline{B}$ at the point P for the current flow shown (hint: the wire is made up of three separate sections).



If the incoming and outgoing leads are considered infinitely long, the contribution of each to the field at P is:

$$H = \frac{I}{4\pi a} (\cos(0) - \cos(90)) = \frac{I}{4\pi a} \dots \text{into the page}$$

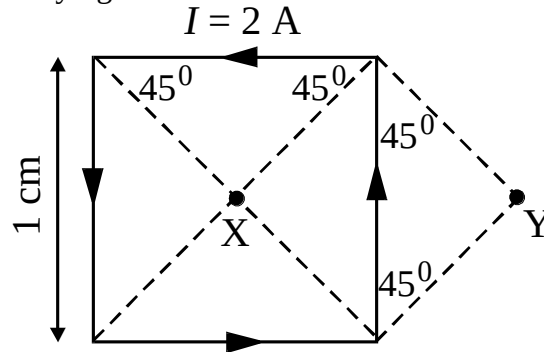
The contribution of the semicircle is $H = \frac{1}{2} \frac{I}{2a} = \frac{I}{4a} \dots \text{into the page}$

The total field at P is thus:

$$\sum H = \frac{I}{4a} \left(1 + \frac{2}{\pi} \right) = 204.6 \text{ A/m}$$

$$B = \mu_0 H = 0.257 \text{ mT}$$

4.6** Calculate B at the points X (**) and Y (***) shown in the diagram below, for a 50 turn square coil, of side 1 cm, carrying a current of 2 A.



At X: Using the solution of the Biot-Savart law for a line segment of finite length, the contribution of one side of one turn of the coil is:

$$H = \frac{I}{4\pi r} (\cos \alpha_1 - \cos \alpha_2) = \frac{I}{2\pi l} (\cos(45^\circ) - \cos(135^\circ)) = \frac{I}{2\pi l} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) = \frac{I}{\sqrt{2}\pi l}$$

The length on one side ' l ' is 1cm, and the radial displacement $r = l/2$

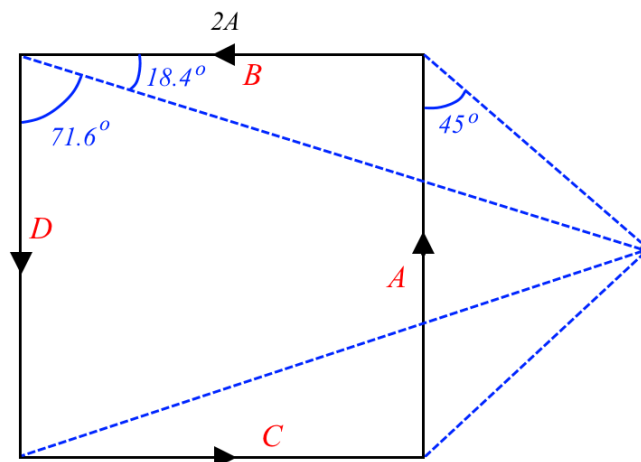
The contribution of each side of the square coil is equal and acts in the same direction (out of the page) – confirm this using the right hand rule.

It follows that each turn must also contribute to the field at X, so the resultant field at the midpoint of the coil is:

$$\sum H = \frac{4NI}{\sqrt{2}\pi l} = 18 \text{ kA/m}$$

$$B = \mu_0 H = 11.3 \text{ mT}$$

At Y: The contribution of each side of the coil is different due to the variation of r , α_1 and α_2 . Redrawing the diagram helps here:



The contribution of side 'A' is the same as before, but acts INTO the page:

$$B_A = \frac{\mu_0 NI}{4\pi r} (\cos \alpha_1 - \cos \alpha_2) = \frac{\mu_0 NI}{2\pi l} (\cos(45^\circ) - \cos(135^\circ)) = \frac{\mu_0 NI}{2\pi l} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) = \frac{\mu_0 NI}{\sqrt{2}\pi l}$$

$$= 2.829 \text{ mT} \otimes$$

The contribution of side 'D' is smaller due to the increased radial distance ($r=3l/2$)

$$B_D = \frac{\mu_0 NI}{4\pi r} (\cos \alpha_1 - \cos \alpha_2) = \frac{\mu_0 NI}{6\pi l} (\cos(71.6^\circ) - \cos(108.4^\circ)) = 0.421 \text{ mT} \odot$$

(Out of page)

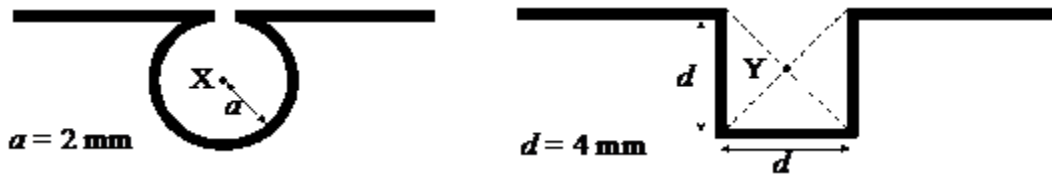
Due to symmetry, the contributions of sides 'B' and 'C' are identical, and also act out of the page:

$$B_B = B_C = \frac{\mu_0 NI}{4\pi r} (\cos \alpha_1 - \cos \alpha_2) = \frac{\mu_0 NI}{2\pi l} (\cos(18.4^\circ) - \cos(45^\circ)) = 0.483 \text{ mT} \odot$$

(Out of page)

The resultant flux density at 'Y' is therefore $B_A - (B_B + B_C + B_D) = 1.442 \text{ mT} \otimes$ into the page

4.7** The following two planar structures consist of a single metal track and may be used as inductors in high frequency circuits. The tracks may be approximated as thin wires.



Assuming that each structure carries a current of 1 mA from left to right and also assuming that the input/output tracks are very long, estimate the magnitude and determine the direction of the magnetic flux density at:

(a) Point X in the first structure.

The loop approximates a circular coil with 1-turn:

$$B_{loop} = \frac{\mu_0 I}{2a} = 3.14 \times 10^{-7} \text{ T} \odot \text{ Out of the page}$$

The input and output tracks may be considered together as an infinitely long wire at a radial distance 'a' from point X

$$B_{tracks} = \frac{\mu_0 I}{2\pi a} = 1 \times 10^{-7} \text{ T} \otimes \text{ Into the page}$$

The resultant flux density at X is therefore $0.214 \mu\text{T} \odot$, directed out of the page

(b) Point Y in the second structure.

The three sides of the square may be treated in the same manner as in question 4.6. Their contribution to the field at 'Y' acts out of the page:

$$B_{sides} = 3 \cdot \frac{\mu_0 NI}{4\pi r} (\cos \alpha_1 - \cos \alpha_2) = 3 \cdot \frac{\mu_0 NI}{2\pi d} (\cos(45^\circ) - \cos(135^\circ)) = 3 \cdot \frac{\mu_0 NI}{2\pi d} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) = 3 \cdot \frac{\mu_0 NI}{\sqrt{2}\pi d}$$

The tracks may again be considered infinitely long, but this time do not reach the midpoint so must be treated as separate wires with $\alpha_1=0^\circ$ and $\alpha_2=45^\circ$

$$B_{tracks} = 2 \cdot \frac{\mu_0 NI}{4\pi r} (\cos \alpha_1 - \cos \alpha_2) = \frac{\mu_0 NI}{\pi d} (\cos(0^\circ) - \cos(45^\circ)) = \frac{\mu_0 NI}{\pi d} \left(1 - \frac{1}{\sqrt{2}} \right) \text{ Into the page}$$

The resultant flux density at 'Y' is the difference = $0.183\mu T \odot$, directed out of the page

4.8** A solenoid has a total of 1000 turns, a radius of 5 mm, a length of 2 cm and carries a current of 0.5 A. Draw a graph of B_z as a function of position along the axis of the solenoid.

From Ampere's law:

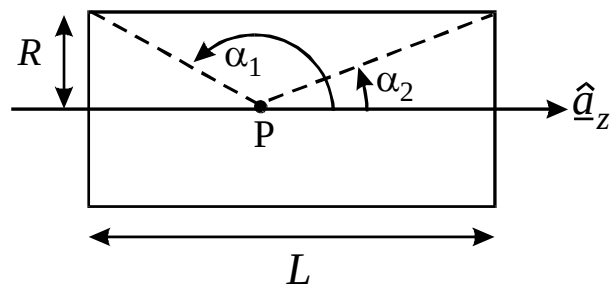
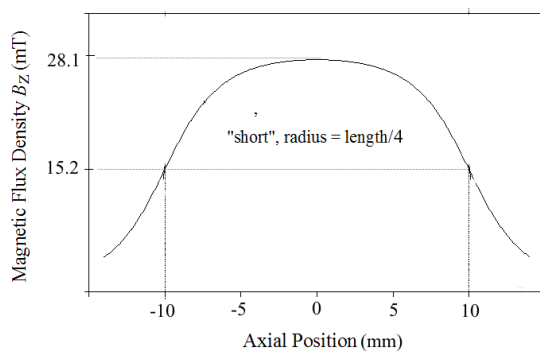
$$B_{inside} = \frac{\mu_0 NI}{l} = 31.4mT$$

$$B_{ends} = \frac{\mu_0 NI}{2l} = 15.7mT$$

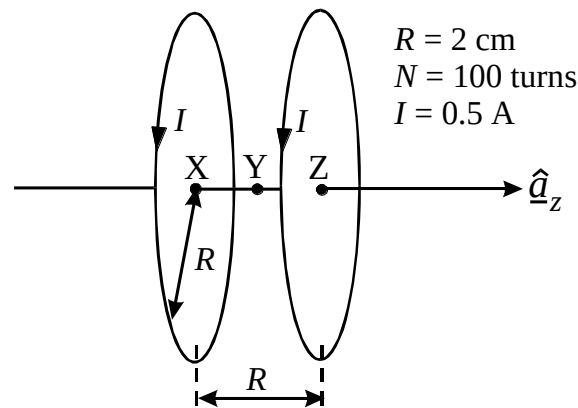
From the Biot Savart Law (exact):

$$B_{inside} = \frac{\mu_0 NI}{2l} (\cos \alpha_1 - \cos \alpha_2) = \frac{\mu_0 NI}{2l} (\cos(26.6^\circ) - \cos(153.4^\circ)) = 28.1mT$$

$$B_{ends} = \frac{\mu_0 NI}{2l} (\cos \alpha_1 - \cos \alpha_2) = \frac{\mu_0 NI}{2l} (\cos(90^\circ) - \cos(166^\circ)) = 15.2mT$$



4.9** For the Helmholtz coils shown, find B_z at the points X, Y and Z.



At Y:

$$B_z = 2 \cdot \frac{\mu_0 N I R^2}{2(R^2 + [R/2]^2)^{3/2}} = \frac{\mu_0 N I}{R} \cdot \left(\frac{4}{5}\right)^{3/2} \approx 0.716 \cdot \frac{\mu_0 N I}{R} = 2.249 \text{ mT} \quad (\text{left to right})$$

At X and Z:

$$B_z = \frac{\mu_0 N I}{2R} + \frac{\mu_0 N I R^2}{2(R^2 + R^2)^{3/2}} = \frac{\mu_0 N I}{R} \cdot \left[\frac{1}{2} + \frac{1}{4\sqrt{2}}\right] = 2.13 \text{ mT}$$

***Also find B_z at a small displacement z from the mid-point Y, and show by Taylor expansion that B_z is constant to order z^4 , thus illustrating the high uniformity of B_z near the mid-point of a pair of Helmholtz coils.

The general expression for the axial flux density due to one coil of the Helmholtz pair is given by

$$B(z) = \frac{\mu_0 N I R^2}{2(R^2 + z^2)^{3/2}}$$

For the coils described above, the value of z at the midpoint is $R/2$. Adding a small displacement ' dz ', the expression for the coil **pair** becomes:

$$B(z) = \frac{\mu_0 N I R^2}{2} \frac{1}{\left(R^2 + \left(dz - \frac{R}{2}\right)^2\right)^{3/2}} + \frac{\mu_0 N I R^2}{2} \frac{1}{\left(R^2 + \left(dz + \frac{R}{2}\right)^2\right)^{3/2}}$$

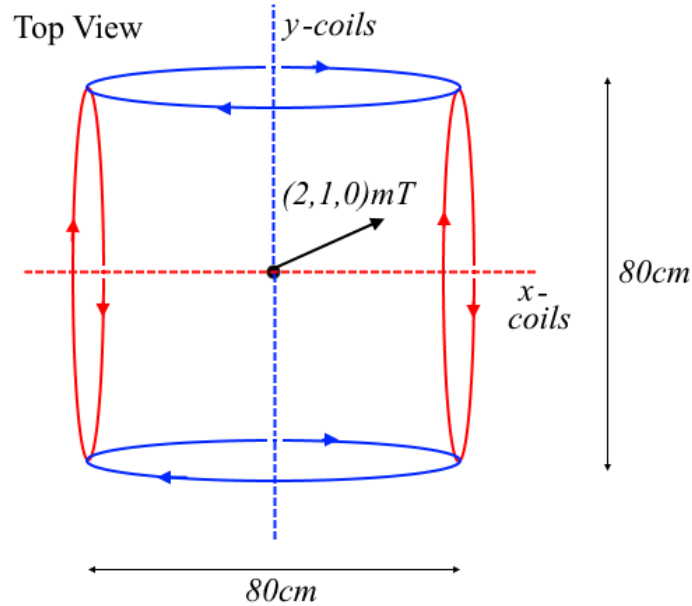
Near the midpoint ($dz \ll R$), the Taylor series expansion is

$$B(z) = \frac{\mu_0 N I R^2}{\left(5R^2/4\right)^{3/2}} \cdot \left[1 + \frac{6}{5} \frac{dz^4}{R^4} \dots\right] = B_M \cdot \left[1 + \frac{6}{5} \frac{dz^4}{R^4} \dots\right] \quad \text{Where } B_M \text{ is the flux density at 'Y'}$$

(Note: dz , dz^2 and dz^3 components are all zero – $B(z)$ is highly uniform about 'Y')

4.10** The magnetic flux density $\underline{B} = (2,1,0)\text{mT}$ at a point needs to be reduced to zero. Show with the aid of a sketch how this can be achieved using two pairs of orthogonal Helmholtz coils, each of radius 40 cm and 500 turns, and calculate the currents required.

Two pairs of coils arranged with their axes directed along x and y can be used to nullify the field as shown:



Rearranging the Helmholtz pair equation, the required current in each coil is given by

$$B_n = \frac{\mu_0 N I_n}{R} \cdot \left(\frac{4}{5}\right)^{3/2} \approx 0.716 \cdot \frac{\mu_0 N I_n}{R}$$

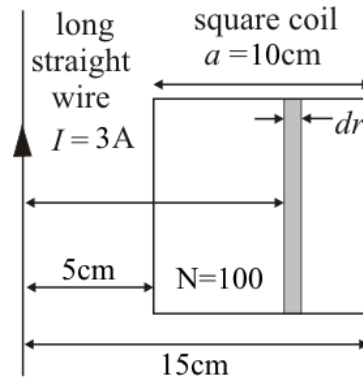
$$\therefore I_n = \frac{B_n R}{0.716 \mu_0 N}$$

If $B_x = 2\text{mT}$, $B_y = 1\text{mT}$, $B_z = 0$:

$$I_x = \frac{B_x R}{0.716 \mu_0 N} = 1.78\text{A}$$

$$I_y = \frac{B_y R}{0.716 \mu_0 N} = 0.89\text{A}$$

4.11** A flat square coil, of side 10 cm and 100 turns, is placed in the plane of a long straight wire carrying a 3 A current. The wire is parallel to the nearest side of the coil, its axis a distance of 10 cm from the centre of the coil. Calculate the flux linked by the coil. (hint: see worked example 5.2 from section 5 of the lecture notes).



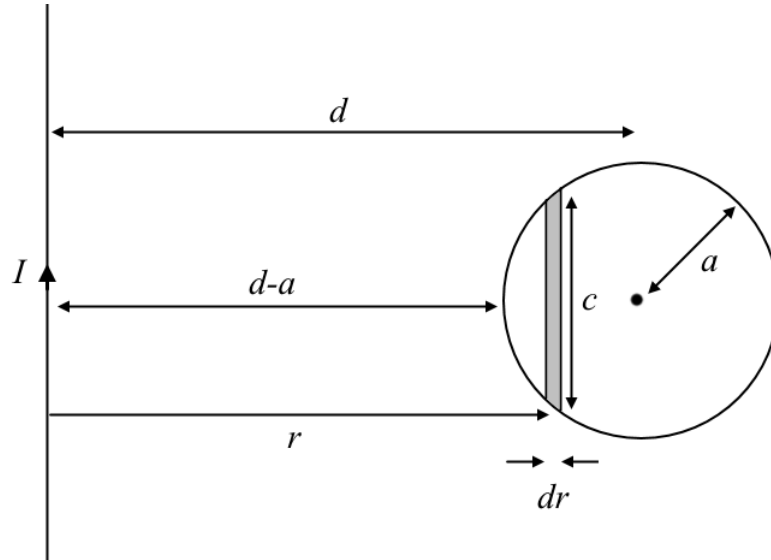
Flux density about a long straight wire: $B(r) = \frac{\mu_0 I}{2\pi r}$

Incremental flux threading the incremental area ' $a \cdot dr$ ': $d\Phi = \frac{\mu_0 I a}{2\pi} \cdot \frac{dr}{r}$

Total flux threading one turn of the coil: $\Phi_1 = \frac{\mu_0 I a}{2\pi} \cdot \int_5^{15\text{cm}} \frac{dr}{r} = \frac{\mu_0 I a}{2\pi} \ln\left(\frac{15}{5}\right) = \frac{\mu_0 I a}{2\pi} \ln(3)$

Total flux threading 100 turns: $\Phi_{TOT} = N \cdot \frac{\mu_0 I a}{2\pi} \ln(3) = 6.59 \mu\text{Wb}$

4.12*** An N turn flat circular coil has radius a and is placed in the plane of a long straight wire carrying a current I . The wire's axis is a distance d ($> a$) from the centre of the coil. Derive a mathematical expression for the flux linked by the coil. (N.B. The resulting integral can't be solved analytically but you may wish to solve it numerically using Mathcad or Matlab; in this instance, assume a 100 turn coil of 1 cm radius, its centre 2 cm from a wire carrying a 5 A current).



Flux density about a long straight wire: $B(r) = \frac{\mu_0 I}{2\pi r}$

Length of chord 'c': $c = 2\sqrt{a^2 - (d-r)^2}$

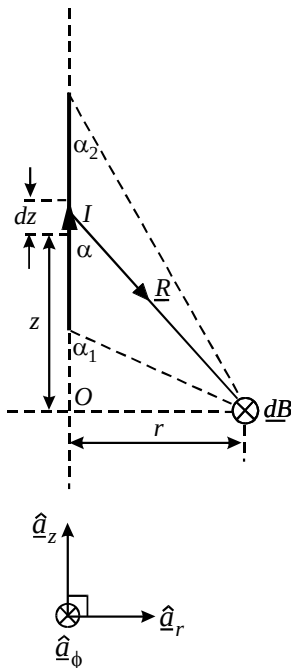
Incremental flux threading the incremental area ' $c \cdot dr$ ': $d\Phi = \frac{\mu_0 I}{\pi} \cdot \frac{\sqrt{a^2 - (d-r)^2}}{r} dr$

Total flux threading N turns of the coil: $\Phi_{TOT} = \frac{\mu_0 NI}{\pi} \cdot \int_{d-a}^{d+a} \frac{\sqrt{a^2 - (d-r)^2}}{r} dr$

(Solve numerically)

4.13** The following derivations using Biot-Savart's law could crop up in the EN2076 examination and are covered in the lecture notes. Do each as an exercise under examination conditions.

(a) Derive $\underline{B}$ outside of a wire of any finite length, and extend to a very long wire.



Consider the flux density $\underline{dB}$ in the direction $\hat{a}_\phi$ (into the page). BS law tells us that

$$\underline{dB} = \frac{\mu_0 I}{4\pi} \frac{d\ell \times \hat{a}_R}{R^2} \rightarrow dB = \frac{\mu_0 I}{4\pi} \frac{dz \sin \alpha}{R^2}$$

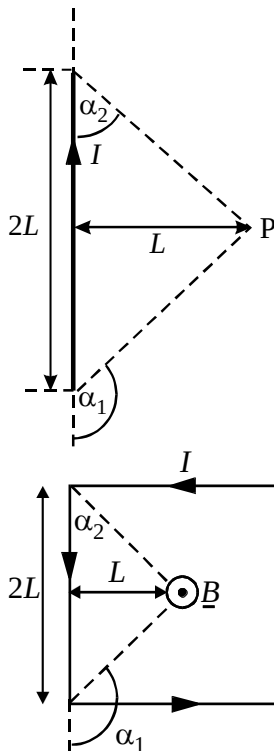
Now change variable from z to α by noting that $z = r \cot \alpha$, i.e. $dz = -r \operatorname{cosec}^2 \alpha d\alpha$

$$\text{Also } R^2 = r^2 + z^2 = r^2(1 + \cot^2 \alpha) = r^2 \operatorname{cosec}^2 \alpha$$

$$\therefore dB = -\frac{\mu_0 I}{4\pi r} \sin \alpha d\alpha \rightarrow B = -\frac{\mu_0 I}{4\pi r} \int_{\alpha_1}^{\alpha_2} \sin \alpha d\alpha$$

$$\text{The final result is } \underline{B} = \frac{\mu_0 I}{4\pi r} (\cos \alpha_2 - \cos \alpha_1) \hat{a}_\phi$$

(b) Derive $\underline{B}$ at the centre of a square coil.



Here $\alpha_1 = 3\pi/4$ (i.e. 135°), $\alpha_2 = \pi/4$ and $r = L$.

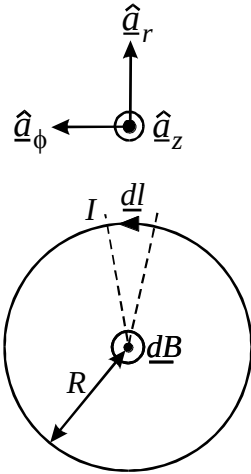
$$\begin{aligned} \therefore \underline{B} &= \frac{\mu_0 I}{4\pi L} \left(\cos \frac{\pi}{4} - \cos \frac{3\pi}{4} \right) \hat{a}_\phi = \frac{\mu_0 I}{4\pi L} \left(\frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}} \right) \right) \hat{a}_\phi \\ &= \frac{\mu_0 I}{\sqrt{2} \cdot 2\pi L} \hat{a}_\phi \quad (\text{points into the page}) \end{aligned}$$

i.e. $\sqrt{2}$ times smaller than the result for a long wire.

Applying to the square coil:

$$B_{\text{mid}} = 4NB = \frac{\sqrt{2} \cdot N\mu_0 I}{\pi L}$$

(c) Derive $\underline{B}$ at any point on the axis of a flat circular coil.

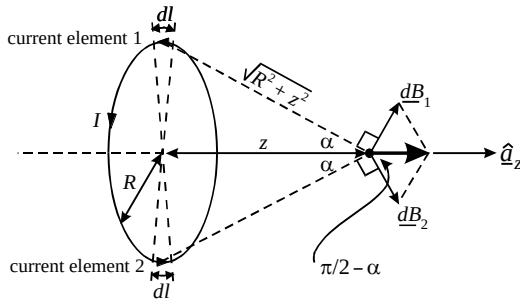


$$d\underline{B} = \frac{\mu_0 I}{4\pi} \frac{d\ell}{R^2} \underline{\hat{a}}_z$$

This has to be integrated over the whole circuit (i.e. a circle of radius R) to find the resultant $\underline{B}$, which points along $\underline{\hat{a}}_z$ (i.e. it is **axial**) when the current I is in the direction of $\underline{\hat{a}}_\phi$:

$$\rightarrow B_z = \frac{\mu_0 I}{4\pi R^2} \int_{\text{circle}} d\ell = \frac{\mu_0 I}{4\pi R^2} \cdot 2\pi R \rightarrow \underline{B} = \frac{\mu_0 I}{2R} \underline{\hat{a}}_z$$

(d) Derive $\underline{B}$ at any point on the axis of a pair of Helmholtz coils.



First, consider two current elements on opposite sides of a coil.

Applying BS law:

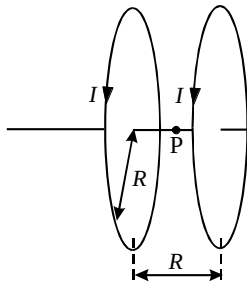
$$dB_1 = dB_2 = \frac{\mu_0 NI}{4\pi} \frac{d\ell}{R^2 + z^2}$$

The resultant field $d\underline{B}$ of the two elements is along $\underline{\hat{a}}_z$ (**axial**) for I along $\underline{\hat{a}}_\phi$.

$$dB_z = (dB_1 + dB_2) \sin \alpha = \frac{\mu_0 NI}{2\pi} \frac{d\ell}{R^2 + z^2} \cdot \frac{R}{(R^2 + z^2)^{1/2}}$$

$$\therefore B_z = \frac{\mu_0 NI}{2\pi} \cdot \frac{R}{(R^2 + z^2)^{3/2}} \cdot \int_{\text{half coil}} d\ell = \frac{\mu_0 NI}{2\pi} \cdot \frac{R}{(R^2 + z^2)^{3/2}} \cdot \pi R = \frac{\mu_0 NIR^2}{2(R^2 + z^2)^{3/2}}$$

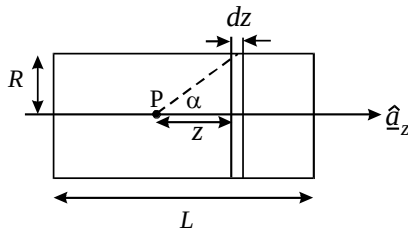
Helmholtz coils are two identical N -turn coils on a common axis with separation equal to their radii R . The same current I is passed through both in the same sense. **They have the useful property of having a highly uniform $\underline{B}$ -field around their mid-point.** Derive an expression for B_z at their mid-point (where $z = \pm R/2$).



Assume that the currents flow in the direction of $\underline{\hat{a}}_\phi$, so that the resultant $\underline{B}$ at the mid-point P is $2B\underline{\hat{a}}_z$ (i.e. it is **axial**), i.e. $B_z = 2B$, where B is the magnitude of the field produced by each coil at P .

$$\therefore B_z = 2 \cdot \frac{\mu_0 NIR^2}{2(R^2 + [R/2]^2)^{3/2}} = \frac{\mu_0 NI}{R} \cdot \left(\frac{4}{5}\right)^{3/2} \approx 0.716 \cdot \frac{\mu_0 NI}{R}$$

(e) Derive $\underline{B}$ at any point on the axis of a solenoid (inside or out).

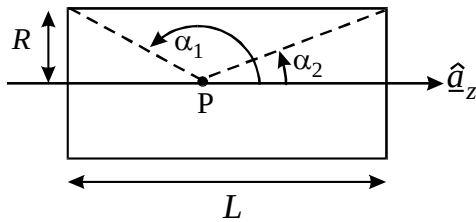


We can think of the solenoid as being made up of an infinite number of current loops. We'll first find the axial field dB_z at a point due to a current loop of thickness dz , then integrate to find the total field B_z .

Using the result of a circular coil, $dB_z = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}} \cdot \frac{N}{L} dz$

Now change variables from distance z to angle α using $z = R \cot \alpha$

$$\therefore dz = -R \operatorname{cosec}^2 \alpha d\alpha \quad \text{and} \quad (R^2 + z^2)^{3/2} = R^3 \operatorname{cosec}^3 \alpha$$



$$\therefore dB_z = -\frac{\mu_0 I N}{2L} \cdot \sin \alpha d\alpha$$

$$\rightarrow B_z = -\frac{\mu_0 I N}{2L} \int_{\alpha_1}^{\alpha_2} \sin \alpha d\alpha$$

The result for the axial field at P is $B_z = \frac{\mu_0 N I}{2L} (\cos \alpha_2 - \cos \alpha_1)$

where the angles α_1 and α_2 are defined in the above figure and depend on the position of point P.

Problems 3 - Electric potential and Laplace's equation

3.1* The electric field $\underline{E}$ in a region of space is found to have a uniform magnitude of 5.3 kV/m directed along the y axis. The electric potential ϕ at the origin (0, 0, 0) is found to be 4.1 kV. Derive a general expression for the electric potential $\phi(x, y, z)$ at the point with position vector (x, y, z) . Hence, find the voltage between the points (4, -2, 0) m and (1, 1, 1) m.

$$\underline{E} = (E_x, E_y, E_z) = (0, 5.3, 0) \text{ kV/m}$$

$$\therefore \frac{\partial \phi}{\partial x} = 0, \quad \frac{\partial \phi}{\partial z} = 0$$

From Laplace's equation, we have $\nabla^2 \phi = \frac{\partial^2 \phi}{\partial y^2} = 0$

Integrating once: $\frac{\partial \phi}{\partial y} = -E_y = -5.3 \text{ kV/m}$

Integrating again: $\phi(x, y, z) = -E_y y + \phi_0$

Inserting boundary conditions $\phi(0, 0, 0) = -E_y \cdot 0 + \phi_0 = \phi_0 = 4.1 \text{ kV}$

General expression: $\phi(x, y, z) = -5.3y + 4.1 \text{ kV}$

$$\phi(4, -2, 0) = -5.3 \cdot -2 + 4.1 = 14.7 \text{ kV}$$

$$\phi(1, 1, 1) = -5.3 \cdot 1 + 4.1 = -1.2 \text{ kV}$$

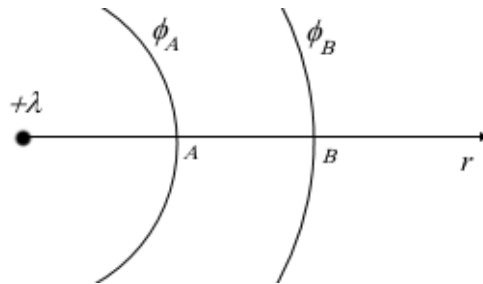
$$V = \phi(4, -2, 0) - \phi(1, 1, 1) = 15.9 \text{ kV}$$

3.2* If the electric potential ϕ at a point with position vector (x, y, z) is $\phi = 2x^3 - z + 4$, use $\underline{E} = -\nabla \phi$ to derive all components of the electric field $\underline{E}$ at $(1, -1, 3)$ m.

$$\underline{E} = -\nabla \phi = -\left(\frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y}, \frac{\partial \phi}{\partial z}\right) = (-6x, 0, 1)$$

Therefore $\underline{E}(1, -1, 3) = (-6, 0, 1) \text{ V/m}$

3.3* A line charge with $\lambda = +1.5 \text{ nC/m}$ sits in free space and is aligned parallel to the z axis, with its axis at the origin (0, 0, 0). Point A is at (1, 0, 0) mm and point B at (2, 0, 0) mm. Calculate the voltage (i.e. potential difference) between points A and B.



$$V_{AB} = \phi_A - \phi_B = -\int_B^A E dr = -\frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{A}{B}\right) = 18.7 \text{ V}$$

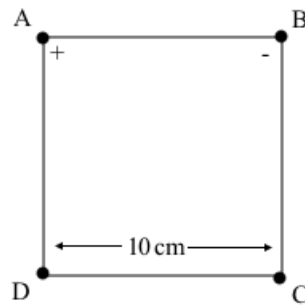
3.4** Repeat problem 3.3 for the voltage between points C (1, 1, 0) mm and D (1, 2, 0) mm.

$$r_C = \sqrt{1^2 + 1^2 + 0} = \sqrt{2}$$

$$r_D = \sqrt{1^2 + 2^2 + 0} = \sqrt{5}$$

$$V_{AB} = \phi_C - \phi_D = -\int_D^C E dr = -\frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{C}{D}\right) = 12.4V$$

3.5** Points ABCD form the corners of a square of side 10 cm, sitting in free space. A free line charge of +10 nC/m is fixed at point A and a line charge of -10 nC/m is fixed at point B. Find the voltage between points D and C, and hence find how much work is done moving a +1 nC point charge between C and D.



The general solution for the electric potential around any one line charge is:

$$\phi = -\frac{\lambda}{2\pi\epsilon_0} \ln(r) + \phi_0, \text{ and taking } d=10\text{cm},$$

$$r_{AD} = r_{BC} = d$$

$$r_{BD} = r_{AC} = d\sqrt{2}$$

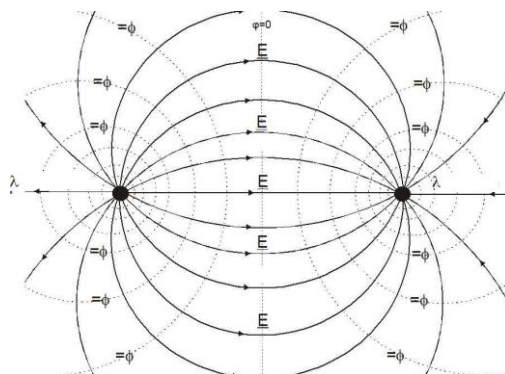
$$\phi_D = -\frac{\lambda}{2\pi\epsilon_0} (\ln(d) - \ln(d\sqrt{2})) + 2\phi_0 = \frac{\lambda}{2\pi\epsilon_0} \ln(\sqrt{2}) + 2\phi_0$$

$$\phi_C = -\frac{\lambda}{2\pi\epsilon_0} (\ln(d\sqrt{2}) - \ln(d)) + 2\phi_0 = -\frac{\lambda}{2\pi\epsilon_0} \ln(\sqrt{2}) + 2\phi_0$$

$$\therefore V_{DC} = \phi_D - \phi_C = \frac{\lambda}{\pi\epsilon_0} \ln(\sqrt{2}) = 124.6V$$

$$W = qV = 124nJ$$

3.6** Sketch the lines of $\underline{E}$ and equipotential surfaces of a line dipole.

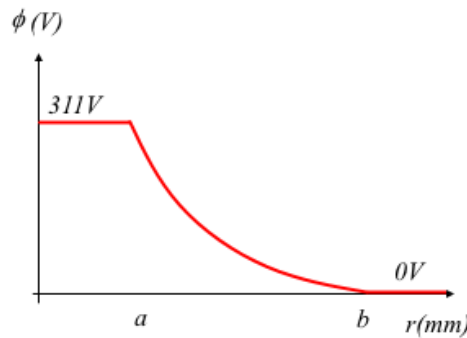


3.7** Calculate the electric potential of the inner conductor of the coaxial line of problem 2.11, and hence find the cable voltage and the cable capacitance. Also calculate the electric potential at the point mid-way between the inner and outer conductors. Plot a graph of the potential variation as a function of radial distance r from the axis of the line.

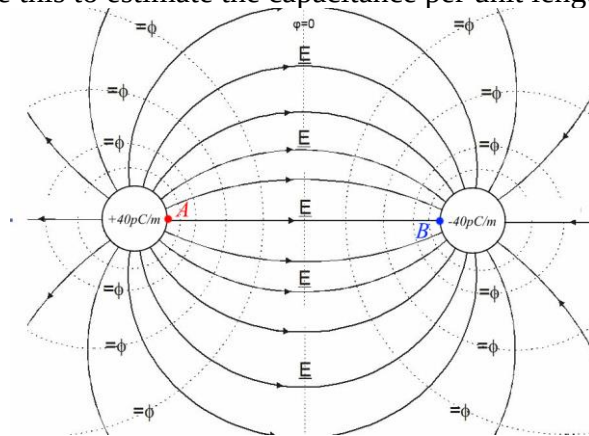
$$V = \phi_a - \phi_b = \frac{\lambda}{2\pi\epsilon_r\epsilon_0} \ln\left(\frac{b}{a}\right) = 311V$$

If b is earthed, $\phi_b = 0$ and thus $\phi_a = 311V$

$$V_m = \phi_m - \phi_b = \frac{\lambda}{2\pi\epsilon_r\epsilon_0} \ln\left(\frac{b}{m}\right) = 118.6V$$



3.8** Two long cylindrical metal wires, each of radius 3 mm, are aligned parallel with their axes a distance of 10 cm apart. One acquires a free charge of +40 pC/m, the other – 40 pC/m. Sketch the lines of $\underline{E}$ and equipotential surfaces in the free space around them. Estimate the electric potential ϕ on the surfaces of each wire. Hence, estimate the voltage between them (i.e. the line voltage) and use this to estimate the capacitance per unit length of this twin wire line.



If $a=3mm$, $D=100mm$:

$$\phi_A = -\frac{\lambda}{2\pi\epsilon_0} (\ln(a) - \ln(D-a)) + 2\phi_0 = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{D-a}{a}\right) + 2\phi_0$$

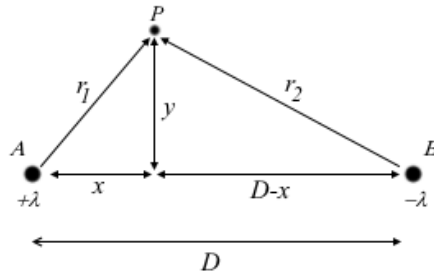
$$\phi_B = -\frac{\lambda}{2\pi\epsilon_0} (-\ln(a) + \ln(D-a)) + 2\phi_0 = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{a}{D-a}\right) + 2\phi_0$$

$$\therefore V_{AB} = \phi_A - \phi_B = \frac{\lambda}{\pi\epsilon_0} \ln\left(\frac{D-a}{a}\right) = 5V$$

Assuming a midpoint potential of 0V (symmetry), we conclude that $\phi_A = +2.5V$ and $\phi_B = -2.5V$

The capacitance of the twin wire is then $C' = \frac{\lambda}{V} = 8 \text{ pF} / \text{m}$

3.9*** Derive a general formula for the electric potential at any point around a line dipole. Hence, use $\underline{E} = -\nabla\phi$ to derive all components of the associated electric field and check your answer by comparing with that obtained in problem 2.10.



From 2.10,

$$\overline{E_x} = \frac{\lambda}{2\pi\epsilon} \cdot \left[\frac{x}{x^2 + y^2} + \frac{(D-x)}{(D-x)^2 + y^2} \right] \hat{a}_x, \quad \overline{E_y} = \frac{\lambda}{2\pi\epsilon} \cdot \left[\frac{y}{x^2 + y^2} - \frac{y}{(D-x)^2 + y^2} \right] \hat{a}_y, \quad \overline{E_z} = 0$$

Using the figure above, the potential at point 'P' is given by:

$$\phi_P = -\frac{\lambda}{2\pi\epsilon} (\ln(r_1) - \ln(r_2)) + \phi_0 \quad (\text{The constant is undefined as no boundary condition present})$$

$$\phi_P = -\frac{\lambda}{2\pi\epsilon} \left(\ln(\sqrt{x^2 + y^2}) - \ln(\sqrt{(D-x)^2 + y^2}) \right) + \phi_0$$

$$= \frac{\lambda}{4\pi\epsilon} \left(\ln((D-x)^2 + y^2) - \ln(x^2 + y^2) \right) + \phi_0$$

$$\underline{E_x} = -\frac{\partial\phi}{\partial x} \hat{a}_x = -\frac{\lambda}{4\pi\epsilon} \left(-\frac{2(D-x)}{(D-x)^2 + y^2} - \frac{2x}{x^2 + y^2} \right) \hat{a}_x = \frac{\lambda}{2\pi\epsilon} \cdot \left[\frac{x}{x^2 + y^2} + \frac{(D-x)}{(D-x)^2 + y^2} \right] \hat{a}_x$$

$$\underline{E_y} = -\frac{\partial\phi}{\partial y} \hat{a}_y = -\frac{\lambda}{4\pi\epsilon} \left(\frac{2y}{(D-x)^2 + y^2} - \frac{2y}{x^2 + y^2} \right) \hat{a}_y = \frac{\lambda}{2\pi\epsilon} \cdot \left[\frac{y}{x^2 + y^2} - \frac{y}{(D-x)^2 + y^2} \right] \hat{a}_y$$

$$\underline{E_z} = -\frac{\partial\phi}{\partial z} \hat{a}_z = 0$$

The x, y and z components are thus consistent with 2.10

3.10** The following derivations could crop up in the EN2076 examination and are covered in the lecture notes. Do each as an exercise under examination conditions.

(a) Show that the solution to Laplace's equation for a uniform field is $\phi = -E x + \phi_0$

$$\nabla^2 \phi = \frac{d^2 \phi}{dx^2} = 0$$

$$\text{Integrating twice} \rightarrow \frac{d\phi}{dx} = A \rightarrow \phi = Ax + \phi_0 \quad (A, \phi_0 \text{ are constants})$$

To find constants A and ϕ_0 we insert the **boundary conditions**, i.e. values that ϕ takes at the boundaries, which are often conducting.

$$\text{In this 1-D example, } \underline{E} = -\underline{\nabla}\phi = -\frac{d\phi}{dx} \underline{\hat{x}} = -A \underline{\hat{x}} \rightarrow A = -E_0$$

$$\rightarrow \phi(x) = -E_0 x + \phi_0$$

(b) Show that the solution to Laplace's equation for a line charge is $\phi = -\frac{\lambda}{2\pi\epsilon_r\epsilon_0} \ln r + \phi_0$.

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \rightarrow \frac{1}{r} \frac{d}{dr} \left(r \frac{d\phi}{dr} \right) = 0$$

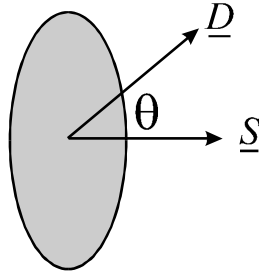
$$\text{Integrating twice} \rightarrow r \frac{d\phi}{dr} = A \rightarrow \phi = A \ln r + \phi_0 \quad (A, \phi_0 \text{ constants})$$

$$\text{The electric field is } \underline{E} = -\underline{\nabla}\phi = -\frac{d\phi}{dr} \underline{\hat{r}} = -\frac{A}{r} \underline{\hat{r}} = E_r \underline{\hat{r}}$$

$$\text{From GL, } E_r = \frac{\lambda}{2\pi\epsilon_r\epsilon_0 r} \rightarrow A = -\frac{\lambda}{2\pi\epsilon_r\epsilon_0} \rightarrow \phi(r) = -\frac{\lambda}{2\pi\epsilon_r\epsilon_0} \ln r + \phi_0$$

Problems 2 - Gauss' law for calculating electric flux density

2.1* A circular plane area (i.e. a disc) of diameter 10 cm is placed with its normal at an angle of 20° to a uniform $\underline{D}$ field of magnitude 0.001 C/m^2 . Calculate the electric flux Φ_e passing through its plane. What happens to this flux as the plane of the disc is rotated?



$$\Phi_e = \underline{D} \bullet \underline{S} = DS \cos \theta = 7.38 \mu\text{C}$$

As θ varies from 0 - 180° , the flux changes from $+7.85 \mu\text{C}$ through to $-7.85 \mu\text{C}$

2.2* A point charge of $+2 \mu\text{C}$ is placed at position vector $(1,0,0)\text{m}$, $-3 \mu\text{C}$ at $(1,1,1)\text{m}$ and $+5 \mu\text{C}$ at $(3,3,0)\text{m}$. Use Gauss' law to find the flux of $\underline{D}$ through the surface of a sphere centred at the origin if its radius is (a) 0.5 m , (b) 2 m , and (c) 5 m ? What is the net flux of $\underline{D}$ outwards through the surfaces of a cube that encloses all of the charges?

$$\begin{aligned} q_1 (1,0,0) &= +2\mu\text{C} & r_1 &= 1\text{m} \\ q_2 (1,1,1) &= -3\mu\text{C} & r_2 &= 1.7\text{m} \\ q_3 (3,3,0) &= +5\mu\text{C} & r_3 &= 4.2\text{m} \end{aligned}$$

GS1 encloses zero charge ($Q_{\text{net}} = 0\text{C}$)

GS2 encloses q_1 and q_2 ($Q_{\text{net}} = -1\mu\text{C}$)

GS3 encloses all charges ($Q_{\text{net}} = +4\mu\text{C}$)

2.3** A free charge of $+10 \text{ nC}$ is deposited onto a thin, flat metal sheet, of area 1 cm^2 . Assuming that the charge spreads out uniformly and ignoring charge crowding at the sheet edges, calculate the free surface charge density, and hence the $\underline{E}$ field magnitude, on the flat surfaces of the sheet. (Hint: the sheet has two flat surfaces). Also sketch the $\underline{E}$ field distribution around the sheet.

With even charge distribution on both faces, $\sigma = 50 \mu\text{C/m}^2$

$$E = \frac{D}{\epsilon_0} = \frac{\sigma}{\epsilon_0} = 5.65 \text{ MV/m}$$

2.4** The charged metal sheet of problem 2.3 is placed parallel, and very near to, an identical earthed metal sheet. Sketch the resulting $\underline{E}$ field distribution and calculate the magnitude of $\underline{E}$ in the free space between the two plates.

The presence of the positively charged sheet induces a negative charge on the earthed sheet. Electrostatic attraction causes the positive charge to redistribute to the face closest to the parallel sheet, meaning $\sigma = 100\mu\text{C}/\text{m}^2$

$$E = \frac{D}{\epsilon_0} = \frac{\sigma}{\epsilon_0} = 11.3\text{MV} / \text{m}$$

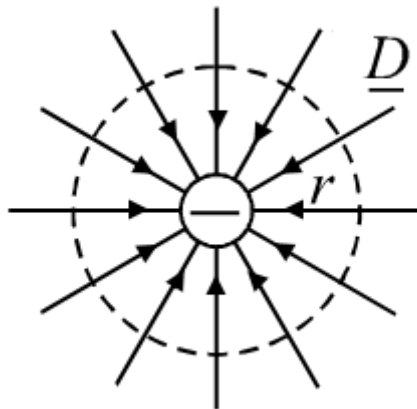
2.5* Calculate the magnitudes of $\underline{E}$ and $\underline{D}$ at distances of 1 mm and 3 mm from a line charge, of free charge density $-8\text{ nC}/\text{m}$, sitting in free space (i.e. a vacuum). Indicate the directions of the fields with a sketch. Also sketch the full field line distributions around the line charge.

$$D = \frac{\lambda}{2\pi r} \quad E = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$\text{for } r=1\text{mm} \quad D=-1.27\mu\text{C}/\text{m}^2 \quad E=-143.8\text{kV}/\text{m}$$

$$\text{for } r=3\text{mm} \quad D=-0.424\mu\text{C}/\text{m}^2 \quad E=-47.9\text{kV}/\text{m}$$

(Negative charge acts as a sink of E-field)



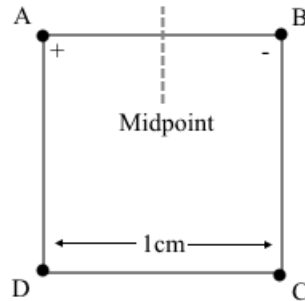
2.6* Repeat the calculation of problem 2.5, this time assuming the line charge to be embedded entirely within in a dielectric material, of relative permittivity 3.8.

$$\text{for } r=1\text{mm} \quad D=-1.27\mu\text{C}/\text{m}^2 \quad E=-37.8\text{kV}/\text{m}$$

$$\text{for } r=3\text{mm} \quad D=-0.424\mu\text{C}/\text{m}^2 \quad E=-12.6\text{kV}/\text{m}$$

(D is unchanged as it is independent of the dielectric material properties - only E is affected)

2.7** ABCD is a square of side 1 cm. A $+0.3 \text{ nC/m}$ line charge is placed with its axis at point A and a second -0.3 nC/m line charge placed at point B, parallel to the first. The lines sit in free space. Sketch the $\underline{E}$ field distribution around this **line dipole**. Calculate the magnitude of $\underline{E}$ at the mid-point between A and B, at C, and also at D. Indicate the directions of $\underline{E}$ at each of these points on your field line plot.



At the midpoint between A and B, the electric field contribution of each dipole act in the same direction (they are additive)

$$D = D_1 + D_2 = 2 \cdot \frac{\lambda}{2\pi \cdot d/2} = \frac{2\lambda}{\pi d} = 19.1 \text{ nC/m}^2 \quad (\rightarrow)$$

The electric field $E = \frac{D}{\epsilon_r \epsilon_0} = \frac{2\lambda}{\pi \epsilon_r \epsilon_0 d} = 2.16 \text{ kV/m} \quad (\text{direction } \rightarrow)$

At 'C', the field contributions are in different directions:

The field due to the charge at 'A' acts in the ↘ direction

The field due to the charge at 'B' acts in the ↑ direction

The radial distances are also different: $r_{CA} = \sqrt{2}$
 $r_{CB} = 1$

E- field contributions: $E_{CA} = \frac{\lambda_A}{\pi \epsilon_r \epsilon_0 r_{CA}} = 381.3 \text{ V/m}$ ↘ $E_{CB} = \frac{\lambda_B}{\pi \epsilon_r \epsilon_0 r_{CB}} = 539.2 \text{ V/m}$ ↑

Resolving E_{CA} into vertical and horizontal components:

$$E_{CA\downarrow} = 381.3 \cos(45^\circ) = 269.6 \text{ V/m}$$

$$E_{CA\rightarrow} = 381.3 \sin(45^\circ) = 269.6 \text{ V/m}$$

The vertical components of E_{CA} and E_{CB} act in opposite directions (subtract). The components of E_C therefore resolve to

$$E_{C\uparrow} = 539.2 - 269.6 = 269.6 \text{ V/m}$$

$$E_{C\rightarrow} = 269.6 \text{ V/m}$$

Vector addition gives: $E_C = 381.3 \text{ V/m}$ ↗

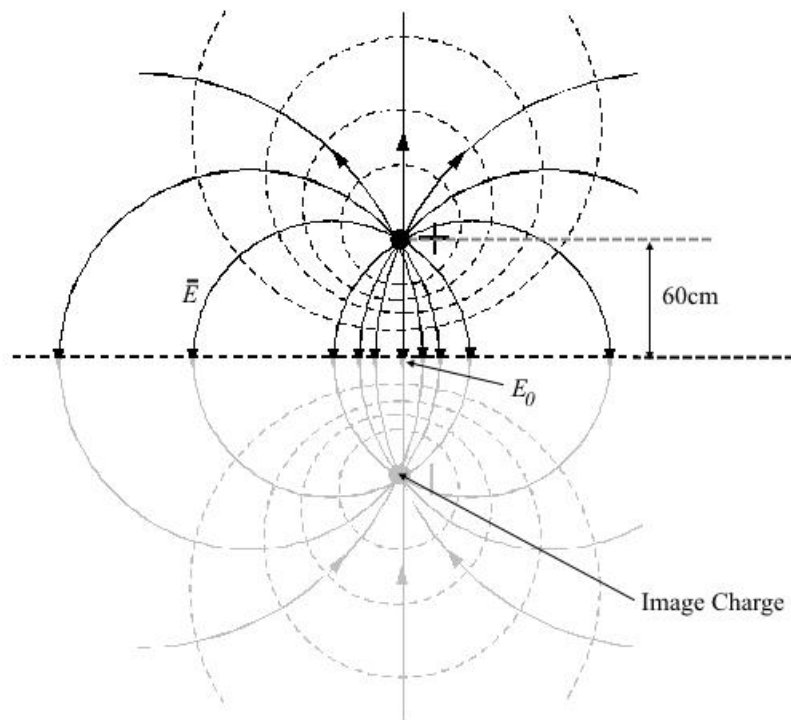
By the same process, we get $E_D = 381.3 \text{ V/m}$ ↙

2.8** Calculate the force per unit length acting on the lines of charge of problem 2.7.

If the force acting on a charged particle in an electric field is $F = Eq$, then the force per unit length will similarly depend on the charge per unit length:

$$F' = E\lambda = \frac{\lambda^2}{2\pi\epsilon_0 d} = 0.162 \mu\text{N} / \text{m}$$

2.9** A $+4.5 \mu\text{C}/\text{m}$ line charge sits in free space with its axis parallel to a large, flat, earthed metal plate, 60 cm away from the plate. For calculating electric fields in and outside of the plate, we use the **method of images**. The $\underline{E}$ field distribution is that of the original charge plus its image charge, equal and opposite and placed an equal distance below the plane defined by the metal surface. Sketch the $\underline{E}$ field distribution in the space outside of the metal. Calculate the $\underline{E}$ field magnitude on the surface of the metal directly below the line charge and the induced surface charge density at this point. Also calculate the force per unit length acting on the line charge.



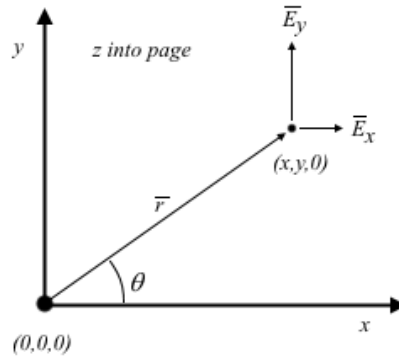
At the earthed metal plate, the electric field magnitude is the same as that at the midpoint of a line dipole. The magnitude and direction of the force is calculated in the same manner. The surface charge density is equal to the flux density at the surface of the metal sheet.

$$E_0 = 2 \cdot \frac{\lambda}{2\pi\epsilon_0 h} = 2.7 \text{ kV/cm}$$

$$F' = E\lambda = \frac{\lambda^2}{2\pi\epsilon_0 (2h)} = 0.303 \text{ N/m} \downarrow$$

$$\sigma = D = \epsilon_0 E_0 = 2.39 \mu\text{C}/\text{m}^2$$

2.10** A long straight line charge (of free charge density $+\lambda$ per unit length) is aligned parallel to the z axis, with its axis at the origin $(0, 0, 0)$. Derive an expression for the electric field $\underline{E} = (E_x, E_y, E_z)$ at a point with position vector $\underline{r} = (x, y, 0)$. *** Extend this result to find a general expression for $\underline{E}$ around a line dipole, aligned parallel to the z axis.



$$r^2 = x^2 + y^2$$

$$\cos(\theta) = \frac{x}{r}$$

$$\sin(\theta) = \frac{y}{r}$$

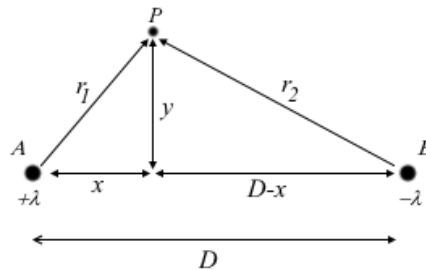
$$\overline{E}_x = \frac{\lambda}{2\pi\epsilon} \cdot \cos(\theta) \hat{a}_x = \frac{\lambda}{2\pi\epsilon} \cdot \frac{x}{r^2} \hat{a}_x = \frac{\lambda}{2\pi\epsilon} \cdot \frac{x}{x^2 + y^2} \hat{a}_x$$

$$\overline{E}_y = \frac{\lambda}{2\pi\epsilon} \cdot \sin(\theta) \hat{a}_y = \frac{\lambda}{2\pi\epsilon} \cdot \frac{y}{r^2} \hat{a}_y = \frac{\lambda}{2\pi\epsilon} \cdot \frac{y}{x^2 + y^2} \hat{a}_y$$

Expressing E as a vector in x, y, z :

$$\underline{\overline{E}} = (E_x, E_y, E_z) = \left(\frac{\lambda}{2\pi\epsilon} \frac{x}{x^2 + y^2}, \frac{\lambda}{2\pi\epsilon} \frac{y}{x^2 + y^2}, 0 \right) = \frac{\lambda}{2\pi\epsilon} \cdot \frac{1}{x^2 + y^2} (x, y, 0) \quad (1)$$

If a line dipole has spacing ' D ', an arbitrary position ' P ' may be expressed in x, y coordinates using the following diagram:



The ' x ' components of electric field both act in the same direction ($\rightarrow\rightarrow$)

The ' y ' components act in opposite directions ($\uparrow\downarrow$)

Using equation (1):

$$\overline{E}_x = \frac{\lambda}{2\pi\epsilon} \cdot \left[\frac{x}{x^2 + y^2} + \frac{(D-x)}{(D-x)^2 + y^2} \right] \hat{a}_x, \quad \overline{E}_y = \frac{\lambda}{2\pi\epsilon} \cdot \left[\frac{y}{x^2 + y^2} - \frac{y}{(D-x)^2 + y^2} \right] \hat{a}_y, \quad \overline{E}_z = 0$$

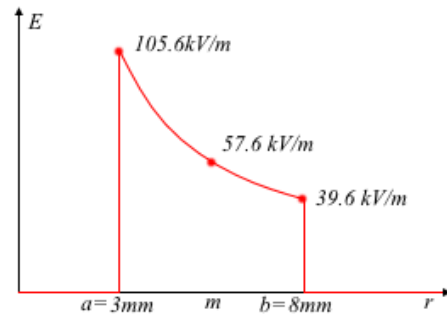
2.11** A coaxial line with $a = 3 \text{ mm}$ and $b = 8 \text{ mm}$ has a dielectric spacer of relative permittivity 2.1. If the free stored charge is $\lambda = \pm 37 \text{ nC/m}$ (with the inner conductor positive and the outer conductor earthed), calculate the $\underline{E}$ field magnitude on the surface of the inner conductor, on the inner surface of the outer conductor, at the point mid-way between the conductors, and hence sketch a graph of $\underline{E}$ field magnitude against radial position r (include points inside and outside of the cable).

$$\bar{E} = \frac{\lambda}{2\pi\epsilon_r\epsilon_0 r}$$

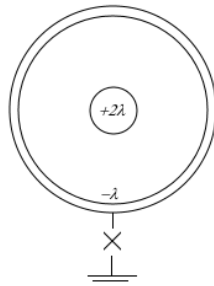
$$E_a = 105.6 \text{ kV/m}$$

$$E_b = 39.6 \text{ kV/m}$$

$$E_m = 57.6 \text{ kV/m}$$



2.12** The earth connection of the coaxial line of problem 2.11 is broken and the charge on the inner conductor is doubled, with the outer conductor remaining isolated. Recalculate the $\underline{E}$ field magnitude on the inner and outer conductors, Assuming the outer conductor has a thickness of 1.5 mm, re-sketch the graph of the $\underline{E}$ field magnitude as a function of radial position r .

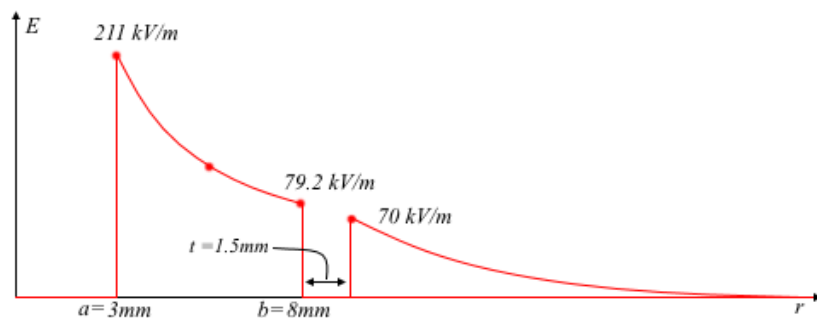


Inside the dielectric spacer, a field doubling occurs due to the doubled charge density

$$E_a = 211.2 \text{ kV/m}$$

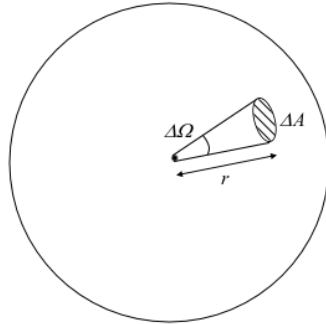
$$E_b = 79.2 \text{ kV/m}$$

Outside the cable, the net charge enclosed is $2\lambda - \lambda = \lambda$, $\bar{E}_{out} = \frac{\lambda}{2\pi\epsilon_0(b+t)} = 70 \text{ kV/m}$



2.13*** Using Coulomb's law as a starting point, derive Gauss' law for the electric flux through an arbitrarily-shaped closed surface (this is a standard proof, using the concept of solid angle, that you will find in any advanced EM textbook – it is non examinable for EN2076).

The solid angle is defined as follows:



$$\Delta\Omega = \frac{\Delta A}{r^2}, \text{ integrating over the surface of a sphere: } \Omega = \int \frac{\Delta A}{r^2} = \frac{4\pi r^2}{r^2} = 4\pi \text{ Steradians}$$

From coulomb's law $E(r) = \frac{q}{4\pi\epsilon_0 r^2}$

The flux through any incremental surface area element is

$$\Delta\Phi = D\Delta A = \epsilon_0 E \cdot r^2 \Delta\Omega = \frac{q}{4\pi} \Delta\Omega$$

Integrating over the solid angle of any closed surface (4π steradians)

$$\Phi = \int_0^{4\pi} \frac{q}{4\pi} \Delta\Omega = \frac{4\pi q}{4\pi} = q$$

(proof of Gauss's law)

2.14** The following derivations could crop up in the EN2076 examination and are covered in the lecture notes. Do each as an exercise under examination conditions.

(a) Use GL to derive $\underline{E}$ at the surface of a charged conductor, i.e. $E = \sigma / \epsilon_r \epsilon_0$.

$$\Phi_e = \oint_S \underline{D} \cdot \underline{dS} = D \cdot A = Q = \sigma A \quad D = \sigma \quad E = \frac{D}{\epsilon_r \epsilon_0} = \frac{\sigma}{\epsilon_r \epsilon_0}$$

(b) Use GL to derive the $\underline{E}$ outside of a free line charge, i.e. $E = \lambda / 2\pi\epsilon_r \epsilon_0 r$.

$$\Phi_e = \oint_S \underline{D} \cdot \underline{dS} = D \cdot 2\pi r l = Q = \lambda l \quad D = \frac{\lambda}{2\pi} \quad E = \frac{D}{\epsilon_r \epsilon_0} = \frac{\lambda}{2\pi\epsilon_r \epsilon_0 r}$$

Problems 10 – Transmission Lines 1

10.1* Sketch the lines of $\underline{E}$ and $\underline{H}$ on the TL structures shown in Section 10.1 of the lecture notes and convince yourself that these fields are TEM. Specify the power flow direction in each case.

10.2* Above what frequency do each of the following lengths of transmission lines become distributed circuits? (a) 2 km long, screened twin cable of phase velocity $0.72c$, (b) 50 cm long, air-spaced coaxial cable, (c) MEMS microstrip line, 1 mm long, with alumina dielectric (of relative permittivity 10.0).

10.3 (refer to your EN1084/EN1085 lecture notes for the relevant theory)** The parallel plate TL (shown in cross-section) can be loosely classed as “wide microstrip”, with approximately uniform EM fields within its dielectric region when $d \ll W$. If at any point along its length the voltage and current amplitudes are v and i , show that the amplitudes of its EM fields are $E \approx v/d$, $H \approx i/W$ (demonstrating the equivalence of the field and circuit description of a TL).



Also show that the capacitance and inductance per unit length are $C \approx \epsilon_r \epsilon_0 W/d$ and $L \approx \mu_0 d/W$, respectively, and use these to derive expressions for the phase velocity of signals $v_p = 1/\sqrt{LC}$ and characteristic impedance $Z_0 = \sqrt{L/C}$. Show how the power carried by the line, calculated in terms of current and voltage amplitudes, relates to the power flux $\underline{S} = \underline{E} \times \underline{H}$ of the EM fields.

10.4(*) (refer to DC's EN2076 lecture notes for the relevant theory)** Repeat problem 10.3 for a coaxial cable of inner and outer radii a and b , respectively. (Hint: when integrating the power flux S to find the total power carried by the cable, take an annular area element of area $dA = 2\pi r dr$).

10.5* (See worked examples 10.2, 10.3) RG58U coaxial cable has a characteristic impedance of 53.5Ω and a dielectric of relative permittivity 2.25. Calculate C and L for this cable. Calculate the wavelength, phase constant, and signal transit time of a 5 kHz signal along a 250 m length of cable.

10.6* (See worked examples 10.3, 10.4) An air-spaced, 300Ω open twin cable is 2.5km long and carries a 1 kV, 50 Hz incident signal. Calculate the current on the cable and hence the rms power flow. How long does after leaving the source does it take the “echo” from the end of the line to return to the source end?

10.7* (See worked example 10.4) A 450Ω transmission line carries 50 kW rms power without reflection. What are the amplitudes of the voltage and current signals along the cable?

10.8 (See worked example 10.5)** A 50Ω stripline connects a 2 GHz, 50Ω microwave source to a 250Ω resistive load, which terminates the stripline. The rms power incident on the load is +35dBm, the substrate has a relative permittivity of 4.1 and the reflected voltage amplitude is $2/3$ times that of the incident voltage amplitude.

- Calculate the voltage and current amplitudes of the incident signal.
- Calculate the voltage and current amplitudes of the reflected signal.
- Calculate the reflected power and the power dissipated by the load (in both W_{rms} and dBm).
- Calculate the amplitudes of the load's voltage and current.
- Explain the relationship between the amplitudes of the load's voltage and current to the amplitudes of the incident and reflected signals (there'll be more of this in Section 11).
- Calculate the phase constant and wavelength of 2 GHz signals along the stripline.
- Write down the phasor forms of the incident and reflected signals, and the load signals.

10.9** The 250Ω resistive load of problem 10.8 is replaced with a 30Ω resistive load and the reflected voltage amplitude is now found to be $-1/4$ times that of the incident voltage amplitude. Repeat parts (a)-(d) for this load, assuming the same incident signal.

10.10 (See worked example 10.6)** Deduce suitable cross-sectional dimensions to fabricate the following transmission lines, filled entirely with a polymer dielectric, of relative permittivity 2.1.

- (a) 25Ω parallel plate transmission line (use the formula you derived in problem 10.3 above).
- (b) 75Ω coaxial cable (use the formula you derived in problem 10.4 above).

10.11** For each of the TLs shown in cross-section in Section 10.1 of the lecture notes, explain qualitatively how the characteristic impedance $Z_0 = \sqrt{L/C}$ can be decreased by changing the cross-sectional dimensions. (Hint: for each structure, think of a way in which C can be increased).

10.12** These derivations sometimes crop up in the EN2076 examination:

- (a) Derive the Telegraphist's equations.
- (b) Use them to derive the wave equations for the voltage and current along a lossless TL.
- (c) Show that the phase velocity of signals along a TEM TL is $v_p = 1/\sqrt{LC} = c/\sqrt{\epsilon_r}$.
- (d) Show that the characteristic impedance of a TEM TL is $Z_0 = \sqrt{L/C}$.

10.13*** Here are some final questions for discussion and for developing a deeper understanding.

(a) If a 1V DC source is connected to a 1Ω resistor then we expect 1A to flow in the circuit that's created. If the wires connecting the source to the load are very long (e.g. 30 km), air-spaced and have zero resistance, then it takes signals 0.1ms to travel along their length. The instant the source is connected to the wires and their 1Ω load, how does the source "know" the 1Ω load is there, in order to provide 1A current? What initial current does it give out? How long is it before this initial current value changes? (All will be revealed in Section 11!)

(b) In TLs like microstrip and coplanar waveguide, some of the electric field energy is in the substrate but a large(ish) fraction of the electric field energy can occupy the air above. Given the statement "signals propagate at the speed of light in the substrate", how is this statement modified to account for "mixed" dielectrics such as the ones described?

(c) A long rectangular tube with a wire down its middle acts as a TEM transmission line, a bit like a coaxial cable. Take the wire away, so we're left with the rectangular tube, then TEM waves are no longer supported. Explain why this is the case, based on what you know about the electric field boundary conditions at a metal surface. Likewise, a coaxial cable supports TEM waves, but remove the inner wire then the cylindrical tube does not. Extend your explanation to account for this.

(d) Why are gas-filled, high voltage TLs used in preference to ones with solid dielectric spacers? What have been the issues with the gases used in the past and what is being done to remedy these issues?

(e) High frequency currents flow within a "skin" near a metal surface, of depth called the "skin depth". Look up the formula for the skin depth of a metal and calculate it for copper, silver and brass at frequencies of 100MHz, 1 GHz and 10 GHz. Use the concept of skin depth to explain why high frequency components are often silver plated. What plating thickness is needed for it to be effective?

(f) What is meant by "dispersion" in a TL and what causes it? Explain the consequences of dispersion by considering what happens to a single square pulse in the time domain after it has propagated some distance along a dispersive TL; extend this to the issues of transmitting digital pulse streams. (Hint: consider the Fourier components of such a square pulse and then the "reconstruction" of the pulse from these components at the far end of the TL). Derive Heaviside's criterion and explain how it helps with dealing with dispersion in practical TLs.

Problems 10 – Outline answers to numerical parts

10.1 The coaxial cable's fields are sketched in section 10.4; for the other structures make sure that $\underline{E}$ points from high to low potential and is perpendicular to the conductors; apply the right hand rule for the direction of $\underline{H}$ relative to current flow, and the power flow direction is in the direction of the cross product of $\underline{E}$ and $\underline{H}$.

10.2 Use the criterion $\ell > \lambda \rightarrow f > v_p / \ell$; (a) 1.08 MHz, (b) 600 MHz, (c) 94.9 GHz

$$10.3 \quad S = EH = vi / Wd, \quad P = \int S dA = S Wd = iv$$

$$10.4 \quad E = v / (r \ln(b/a)), \quad H = i / 2\pi r, \quad S = vi / (2\pi r^2 \ln(b/a)), \quad P = \int S dA = \int_a^b 2\pi r S dr = iv \text{ (again!)}$$

$$10.5 \quad L = Z_0 / v_p = 267 \text{ nHm}^{-1}, \quad C = 1 / Z_0 v_p = 93.9 \text{ pFm}^{-1}, \quad \lambda = v_p / f = 40 \text{ km}, \quad \beta = 1.57 \times 10^{-4} \text{ m}^{-1}$$

transit time $\tau = \ell / v_p = 1.25 \mu\text{s}$

10.6 Current amplitude $I_0^+ = V_0^+ / Z_0 = 3.33 \text{ A}$, rms power flow $P_{\text{in}} = \frac{1}{2} V_0^+ I_0^+ = 1.67 \text{ kW rms}$. Transit time $\tau = \ell / v_p = 8.34 \mu\text{s}$, time taken for echo to return to source is $2\tau = 2\ell / v_p = 16.7 \mu\text{s}$.

$$10.7 \quad \text{Since } P_{\text{in}} = V_0^{+2} / 2Z_0 \rightarrow V_0^+ = \sqrt{2P_{\text{in}}Z_0} = 6.71 \text{ kV}, \quad I_0^+ = V_0^+ / Z_0 = 14.9 \text{ A}.$$

$$10.8 \quad \text{(a) } P_{\text{in}} = +35 \text{ dBm} = 10^{35/10} \text{ mWrms} = 3.16 \text{ W rms}; \quad V_0^+ = \sqrt{2P_{\text{in}}Z_0} = 17.8 \text{ V}, \quad I_0^+ = V_0^+ / Z_0 = 0.356 \text{ A}$$

$$\text{(b) } V_0^- = 0.667 \times 17.8 = 11.9 \text{ V}, \quad I_0^- = -V_0^- / Z_0 = -0.237 \text{ A}$$

$$\text{(c) } P_{\text{ref}} = -\frac{1}{2} V_0^- I_0^- = 1.41 \text{ Wrms}, \quad P_{\text{load}} = P_{\text{in}} - P_{\text{ref}} = 1.76 \text{ Wrms}$$

$$\text{(d) } V_{\text{load}} = \sqrt{2P_{\text{load}}Z_{\text{load}}} = 29.6 \text{ V}, \quad I_{\text{load}} = V_{\text{load}} / Z_{\text{load}} = 0.119 \text{ A}$$

$$\text{(e) } V_{\text{load}} = V_0^+ + V_0^-, \quad I_{\text{load}} = I_0^+ + I_0^- = (V_0^+ - V_0^-) / Z_0$$

$$\text{(f) } \beta = \omega / v_p = 2\pi f \sqrt{\epsilon_r} / c = 84.9 \text{ m}^{-1}, \quad \lambda = 2\pi / \beta = 7.40 \text{ cm}$$

$$\text{(g) Incident signal } v^+(z, t) = 17.8 e^{j(\omega t - \beta z)} \text{ V}, \quad i^+(z, t) = 0.356 e^{j(\omega t - \beta z)} \text{ A}$$

$$\text{Reflected signal } v^-(z, t) = 11.9 e^{j(\omega t + \beta z)} \text{ V}, \quad i^-(z, t) = -0.237 e^{j(\omega t + \beta z)} \text{ A}$$

The load signals are pure AC phasors, not travelling waves, as the load is at a fixed position $z = 0$; hence $v_{\text{load}}(t) = 29.7 e^{j\omega t} \text{ V}$, $i_{\text{load}}(z, t) = 0.119 e^{j\omega t} \text{ A}$.

$$10.9 \quad \text{(a) } P_{\text{in}} = +35 \text{ dBm} = 3.16 \text{ W rms}; \quad V_0^+ = \sqrt{2P_{\text{in}}Z_0} = 17.8 \text{ V}, \quad I_0^+ = V_0^+ / Z_0 = 0.356 \text{ A (as before)}$$

$$\text{(b) } V_0^- = -0.25 \times 17.8 = -4.45 \text{ V}, \quad I_0^- = -V_0^- / Z_0 = +0.0890 \text{ A}$$

$$\text{(c) } P_{\text{ref}} = -\frac{1}{2} V_0^- I_0^- = 0.198 \text{ Wrms}, \quad P_{\text{load}} = P_{\text{in}} - P_{\text{ref}} = 2.96 \text{ Wrms}$$

$$\text{(d) } V_{\text{load}} = \sqrt{2P_{\text{load}}Z_{\text{load}}} = 13.3 \text{ V} = V_0^+ + V_0^-, \quad I_{\text{load}} = V_{\text{load}} / Z_{\text{load}} = 0.445 \text{ A} = I_0^+ + I_0^-$$

10.10 (a) For a parallel plate TL, $Z_0 = \eta d / W$ so $d / W = 0.0962$ for a 25Ω line; (b) for a coaxial cable, $Z_0 = \eta \ln(b/a) / 2\pi$ so $b/a = 6.13$ for a 75Ω cable.

Problems 11 – Transmission Lines 2

11.1* (See worked example 11.1) Find the reflected and load signals (voltage and current) for an incident 100V, 2A DC signal on a 50Ω TL terminated with a (a) matched load, (b) 125Ω load, and (c) 10Ω load. Show that the load power is the difference between incident and transmitted powers.

11.2* (See worked example 11.1) How would the calculations for worked example 11.1 differ for a AC incident signal of amplitudes 100V, 2A?

11.3* (See worked example 11.2) Repeat worked example 11.1 if the TL were to be terminated in (a) an open circuit, (b) a short circuit. Comment on the power flow in these two cases.

11.4 (See worked example 11.3)** A 400V, 2A AC signal travels along a 200Ω TL and arrives at a junction with a 400Ω TL. Calculate the reflected and transmitted powers at the junction.

11.5 (See worked example 11.4)** A 75Ω TL forms a parallel junction with a 50Ω TL and a 100Ω TLs. If a 300V, 4A AC signal is incident along the 75Ω line, calculate the incident, reflected and transmitted powers; hence show that power is conserved.

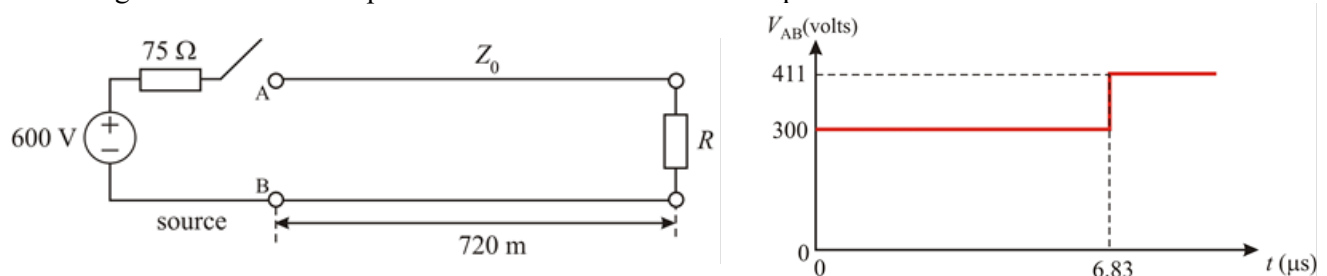
11.6 (See worked example 11.5)** A 100Ω shunt resistor is connected across the conductors of a long 50Ω TL. The TL supports an AC signal of current amplitude 1.2A. Calculate the reflected, transmitted and dissipated powers at the junction with the resistor. Show that power is conserved.

11.7 (See worked examples 11.4, 11.5)** A 200Ω shunt resistor is connected across the junction of Problem 11.5. If the original source is removed and a signal (AC or DC) with power P is sent along the 50Ω line, what % of this incident power is reflected, what % is transmitted along the 100Ω line, what % is transmitted along the 75Ω line, and what % is dissipated in the 100Ω shunt resistor?

11.8* (See worked example 11.6) What matched voltage sources are required to set up the incident signals in Problems 11.1, 11.2, 11.4, 11.5 and 11.6?

11.9 (See worked examples 11.7, 11.8)** A 200Ω twin wire line connects a 50V AC source (of output impedance 200Ω) to the following loads (a) 500Ω , (b) 70Ω , (c) $47k\Omega$, (d) 1Ω . Calculate the power dissipated in each load, expressed as a sum of incident and reflected powers on the twin wire line. What is the maximum power that this source can deliver?

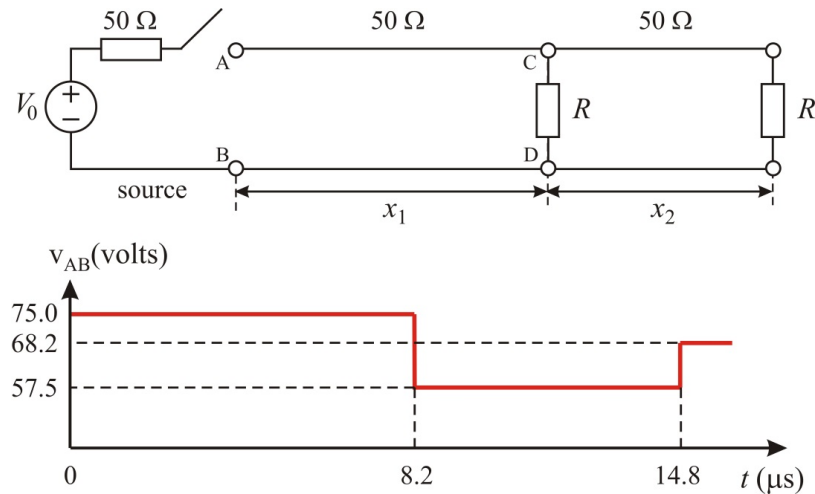
11.10 (see worked example 11.9)** The TL circuit shown connects a 600 V DC voltage source, of output impedance 75Ω , to an unknown load of resistance R . The switch is closed at time $t = 0\mu s$ and the voltage across the TL's input terminals AB is measured and plotted.



Explain the features in the graph and use it to find (a) the phase velocity of the TL, (b) the characteristic impedance Z_0 of the TL, (c) its inductance and capacitance per unit length, (d) the load's reflection coefficient, and (hence) (e) the value of load resistance R .

11.11 (see worked examples 11.9, 11.10)** In Problem 11.10, sketch similar graphs of the current at the TL's terminals AB, the load voltage and the load current, all as functions of time t after the switch has been closed at $t = 0\mu\text{s}$. If the load develops a sudden short circuit fault at $t = 10\mu\text{s}$, append all four graphs to account for this change.

11.12 (see worked examples 11.9, 11.10)** The circuit shown below consists of two air-spaced 50Ω transmission lines, with equal resistors R connected at the positions shown. Also shown is the time dependence of the voltage across the input terminals AB after the switch has been closed (at $t = 0\mu\text{s}$).



- Use the graph to calculate the source voltage V_0 , the lengths x_1 and x_2 , and the load resistance R .
- Sketch similar, fully labelled graphs for the TL current at AB, the voltage across terminals CD and the TL current at CD, all for times in the range $0 \leq t \leq 15\mu\text{s}$.
- Calculate the steady state signals at AB and CD and indicate these on your graphs.

11.13 (see worked example 11.11)** Calculate the VSWR, the maximum and minimum voltages, and the maximum and minimum currents in the standing wave set up when 100W_{rms} RF power at 200MHz travels along a 50Ω TL of phase velocity $0.65c$ terminated with (a) a 50Ω load, (b) an open circuit, (c) a short circuit, (d) a 30Ω load, and (e) a 90Ω load. Sketch labelled graphs of the voltage and current envelopes in each case and comment on the power flow.

11.14*** A DC power supply, of voltage V_S and output impedance Z_S , is switched into a length of transmission line of characteristic impedance Z_0 , terminated with a load Z_L . Assuming that nothing is matched (i.e. $Z_S \neq Z_0 \neq Z_L$), show mathematically that the steady state signal levels (predicted by simple circuit theory) are the result of the summation of the multiple signal reflections on the line.

11.15** The only derivation from Section 11 that sometimes crops up in the EN2076 examination is to show that a load's voltage reflection coefficient is

$$\rho_L = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

11.16*** Here are some final questions for discussion and for developing a deeper understanding.

- We've considered TL transients when a source is switched on and into a TL circuit. Discuss the transients that occur when the source is switched off, so voltage and current amplitudes have to return to 0V , 0A at all points along the TL. Consider various situations (e.g. matched source/TL/load, matched source/TL but unmatched load, etc.).
- Extend the discussion of transients to the propagation of pulses along TL's and the effect that reflections have on the transmission of digital data (i.e. bitstreams).

- (c) Are TL surges problematical (i.e. the transient signal amplitudes that are present before steady state conditions are attained)?
- (d) Write a short MATLAB or Mathcad program that computes the input impedance of a transmission line terminated with a load for any values of Z_0 , Z_L and electrical length βL . Vary βL for a 50Ω line terminated with a 100Ω load, firstly with $\beta L \ll 1$ and secondly with βL in the range 0.1 to 10. Comment on the practical implication of these results for mains (50Hz) TLs and for very high frequency TLs (> 1 GHz).
- (e) What is a stub tuner and how can it be used to match a TL's load in high frequency circuits.
- (f) What is a Smith chart? Discuss its role in aiding high frequency circuit design (this will be good preparation if you intend taking the module EN3082 HF and RF Engineering in Year 3)

Problems 11 - Outline answers to numerical parts

11.1 (a) There is no reflection from a matched load so the load voltage and current are those of the incident wave, i.e. 100V, 2A, and the load power equals the incident power, in this case 200W (note it's DC). (b) The voltage reflection coefficient is +0.429 so the reflected voltage and current are $V_0^- = 42.9\text{V}$, $I_0^- = -0.857\text{A}$, so the reflected power is 36.7W; the load voltage is $100 + 42.9 = 142.9\text{V}$, the load current is $2 - 0.857 = 1.143\text{A}$, so the load power is 163.3W (i.e. $200\text{W} - 36.7\text{W}$). (c) The voltage reflection coefficient is -0.667 so the reflected voltage and current are $V_0^- = -66.7\text{V}$, $I_0^- = 1.33\text{A}$, so the reflected power is 88.9W; the load voltage is $100 - 66.7 = 33.3\text{V}$, the load current is $2 + 1.33 = 3.33\text{A}$, so the load power is 111W (i.e. $200\text{W} - 89\text{W}$).

11.2 The amplitudes of all voltages and currents would be the same but we would halve all powers since for AC we deal with rms values, averaged over a full AC cycle, or integer numbers of AC cycles.

11.3 (a) For an open circuit termination, voltage reflection coefficient is +1 so the reflected voltage and current are $V_0^- = 100\text{V}$, $I_0^- = -2\text{A}$, resulting in a voltage of 200V across the open circuit end but zero current through it; that means the load power is zero, and the powers of the incident and reflected signals are equal and opposite. (b) For a short circuit termination, voltage reflection coefficient is -1 so the reflected voltage and current are $V_0^- = -100\text{V}$, $I_0^- = 2\text{A}$, resulting in zero voltage across the open circuit end but 4A current through it; that means the load power is again zero, and the powers of the incident and reflected signals are again equal and opposite.

11.4 The incident signal amplitudes are $V_1^+ = 400\text{V}$, $I_1^+ = 2\text{A}$ and the incident power is 400Wrms; the voltage reflection coefficient of the junction is +0.333 so the reflected signal amplitudes are $V_1^- = 133.3\text{V}$, $I_1^- = -0.667\text{A}$ and the reflected power is 44.4Wrms; the junction voltage on arrival of the incident signal is $V_J = 400 + 133.3 = 533.3\text{V}$, so the transmitted signal amplitudes are $V_2^+ = 533.3\text{V}$, $I_2^+ = 1.333\text{A}$ and the transmitted power is 355.6W. Note that $355.6 + 44.4 = 400$, so we have conservation of power (which we must always have).

11.5 The incident signal amplitudes are $V_1^+ = 300\text{V}$, $I_1^+ = 4\text{A}$ and the incident power is 600Wrms. The input impedance looking into the junction from the 75Ω line is $50 \parallel 100 = 33.3\Omega$ so the voltage reflection coefficient is -0.385 so the reflected signal amplitudes are $V_1^- = -115.4\text{V}$, $I_1^- = 1.539\text{A}$ and the reflected power is 88.8Wrms; the junction voltage on arrival of the incident signal is $V_J = 300 - 115.4 = 184.6\text{V}$, so the transmitted signal amplitudes on the 50Ω line are $V_2^+ = 184.6\text{V}$, $I_2^+ = 3.692\text{A}$ and the power transmitted along it is 340.8Wrms; the transmitted signal amplitudes on the 100Ω line are $V_3^+ = 184.6\text{V}$, $I_3^+ = 1.846\text{A}$ and the power transmitted along it is 170.4Wrms; since $88.8 + 340.8 + 170.4 = 600$, we have verified that power is conserved.

11.6 The incident signal amplitudes are $V_1^+ = 60\text{V}$, $I_1^+ = 1.2\text{A}$ and the incident power is 36Wrms. The input impedance at the junction is $100 \parallel 50 = 33.3\Omega$ so the voltage reflection coefficient is -0.2 so the reflected signal amplitudes are $V_1^- = -12\text{V}$, $I_1^- = 0.24\text{A}$ and the reflected power is 1.44Wrms; the junction voltage on arrival of the incident signal is $V_J = 60 - 12 = 48\text{V}$, so the transmitted signal amplitudes on the rest of the 50Ω line are $V_2^+ = 48\text{V}$, $I_2^+ = 0.96\text{A}$ and the power transmitted along it

is 23.04Wrms; the shunt resistor voltage and current amplitudes are 48V, 0.48A so the power dissipated in it is 11.52Wrms; since $36 = 1.44 + 23.04 + 11.52$, we have verified that power is conserved.

11.7 It's easiest to let the rms incident power P along the 50Ω line to be 100Wrms, in which case all other powers will compute directly to be % values of P . The incident signal amplitudes along the 50Ω line are therefore $V_1^+ = \sqrt{2PZ_1} = 100\text{V}$, $I_1^+ = 2\text{A}$; the input resistance looking into the junction from the 50Ω line is $75 \parallel 100 \parallel 100 = 75 \parallel 50 = 30\Omega$, so the voltage reflection coefficient is -0.25 and the reflected signal amplitudes are $V_1^- = -25\text{V}$, $I_1^- = 0.5\text{A}$ and the reflected power is 6.25Wrms (i.e. 6.3% of P is reflected at the junction); the junction voltage on arrival of the incident signal is $V_j = 100 - 25 = 75\text{V}$, so the resistor voltage and current are 75V, 0.75A so the power dissipated in it is 28.1Wrms, i.e. 14.1% of P ; the signal launched along the 75Ω line has amplitudes 75V, 1A so its transmitted power is 37.5Wrms, i.e. 37.5% of P ; finally, the signal launched along the 100Ω line has amplitudes 75V, 0.75A so its transmitted power is 28.1Wrms, i.e. 28.1% of P ; since $6.3 + 28.1 + 37.5 + 28.1 = 100$, we have verified that power is conserved.

11.8 In 11.1 we need a 200V DC source with 50Ω output impedance; in 11.2 we need a 200V AC source with 50Ω output impedance; in 11.4 we need an 800V AC source with 200Ω output impedance; in 11.5 we need a 600V AC source with 75Ω output impedance; in 11.6 we need a 120V AC (voltage) with 50Ω output impedance.

11.9 This source will launch a 25V, 0.125A (i.e. 1.56Wrms power) signal along it, which will be totally absorbed by a matched load (i.e. 200Ω), so the maximum source power is 1.56Wrms into a 200Ω load. (a) The voltage reflection coefficient is $+0.375$, the reflected signal amplitudes are 9.38V, -0.0469A , so the reflected power is 0.22Wrms; the load voltage and current are $25 + 9.38 = 34.38\text{V}$, $0.125 - 0.0469 = 0.0781\text{A}$, so the load power is 1.34Wrms (i.e. the difference between incident and reflected powers); (b) The voltage reflection coefficient is -0.482 , the reflected signal amplitudes are -12.04V , -0.0602A , so the reflected power is 0.36Wrms; the load voltage and current are $25 - 12.04 = 12.96\text{V}$, $0.125 + 0.0602 = 0.185\text{A}$, so the load power is 1.20Wrms (i.e. the difference between incident and reflected powers); (c) The voltage reflection coefficient is $+0.992$, the reflected signal amplitudes are 24.79V, -0.124A , so the reflected power is 1.54Wrms; the load voltage and current are $25 + 24.79 = 49.79\text{V}$, $0.125 - 0.124 = 0.001\text{A}$, so the load power is only 0.02Wrms (i.e. the difference between incident and reflected powers) – note this is tending towards an open circuit load; (d) The voltage reflection coefficient is -0.990 , the reflected signal amplitudes are -24.75V , 0.124A , so the reflected power is 1.53Wrms; the load voltage and current are $25 - 24.75 = 0.25\text{V}$, $0.125 + 0.124 = 0.249\text{A}$, so the load power is only 0.03Wrms (i.e. the difference between incident and reflected powers) – note this is tending towards a short circuit load.

11.10 (a) The “echo” from the load arrives back at the source after $6.83\mu\text{s}$, over which time it travels 1440m, so the phase velocity is $2.11 \times 10^8 \text{ m/s}$; (b) the signal launched down the line has amplitude 300V and since this is half the source voltage then we must have $Z_0 = Z_S = 75\Omega$; (c) using the methods developed in Section 10, $C = 1/v_p Z_0 = 63.2 \text{ pF/m}$, $L = Z_0 / v_p = 356 \text{ nH/m}$; (d) the incident voltage is 300V and the reflected voltage is $411 - 300 = +111\text{V}$, so the load's voltage reflection coefficient is $+0.370$; (e) hence $0.37 = (R - 75)/(R + 75)$, so solving this we find $R = 163\Omega$.

11.11 Over to you to draw these graphs.

11.12 (a) The voltage launched down the TL is 75V and it's matched to the source, so the source voltage V_0 must be 150V. The first echo arrives back at the source after $8.2\mu\text{s}$ so the distance x_1 is

covered in a time of $4.1\mu\text{s}$ at the speed of light c , so $x_1 = 1229\text{m}$; likewise, the second “echo” is reflection off the furthest resistor, so x_2 is covered in a time of $(14.8 - 8.2)/2 = 3.3\mu\text{s}$, i.e. $x_2 = 989\text{m}$; the reflection coefficient off junction CD is $-(75.0 - 57.5)/75 = -0.233 = (Z - 50)/(Z + 50)$, where Z is the input impedance of the junction CD, i.e. $Z = R \parallel 50 = 50R/(R + 50)$; we first find $Z = 31.1\Omega$, so that $R = 82.3\Omega$; (b) over to you to draw these graphs; (c) The total resistance of the circuit provided to the source is 91.2Ω , so the total current drawn is 1.65A ; half of this will pass through each R and the voltage across AB and CD will be 67.7V .

11.13 Common to all parts (a)-(e) will be incident voltage and current amplitudes of 100V , 2A ; also, the wavelength on the TL is $\lambda = v_p / f = 0.65c / f = 0.974\text{m}$; (a) there is no reflected signal so maximum and minimum voltage amplitudes are 100V , maximum and minimum currents are 2A , there is no standing wave (constant amplitude everywhere), there’s maximum power transfer to the load and the $\text{VSWR} = 1$; (b) the voltage reflection coefficient is $+1$, the reflected amplitudes are 100V , -2A , the maximum and minimum voltages are 200V and 0V , the maximum and minimum currents are 4A and 0A , we have a perfect standing wave with no net power flow; the VSWR is infinite; (c) the voltage reflection coefficient is -1 , the reflected amplitudes are -100V , 2A , the maximum and minimum voltages are 200V and 0V , the maximum and minimum currents are 4A and 0A , we again have a perfect standing wave with no net power flow; the VSWR is infinite; the difference between the standing wave plot of amplitude envelopes with that of (b) will be that the voltage is zero across the short circuit, whereas it would be 200V across the open circuit of (b); (d) the voltage reflection coefficient is -0.25 , the reflected signal amplitudes are -25V , 0.5A , the maximum and minimum voltages are 125V and 75V , the maximum and minimum currents are 2.5A , 1.5A ; we now have a partial standing wave with net power flow into the load and the load power is 93.8Wrms (i.e. the difference of the 100Wrms incident power and 6.25Wrms reflected power); the $\text{VSWR} = 1.67$; (e) the voltage reflection coefficient is $+0.286$, the reflected signal amplitudes are 28.6V , -0.571A , the maximum and minimum voltages are 128.6V and 71.4V , the maximum and minimum currents are 2.571A , 1.429A ; we now have a partial standing wave with net power flow into the load and the load power is 91.8Wrms (i.e. the difference of the 100Wrms incident power and 8.17Wrms reflected power); the $\text{VSWR} = 1.80$. (To draw the amplitude envelope plots for voltage and current, simply re-draw the plots from Section 11.9 and add the numerical detail covered above).

11.14 The initial surge voltage is $V_0^+ = \frac{Z_0}{Z_S + Z_0} V_S$

The load and source reflection coefficients are $\rho_L = \frac{Z_L - Z_0}{Z_L + Z_0}$, $\rho_S = \frac{Z_S - Z_0}{Z_S + Z_0}$

The limiting value of load voltage is (on summing the multiple voltage reflections)

$$V_L = V_0^+ (1 + \rho_L + \rho_S \rho_L + \rho_S \rho_L^2 + \rho_S^2 \rho_L^2 + \rho_S^2 \rho_L^3 + \rho_S^3 \rho_L^3 + \rho_S^3 \rho_L^4 + \rho_S^4 \rho_L^4 + \rho_S^4 \rho_L^5 + \rho_S^5 \rho_L^5 \dots)$$

$$= V_0^+ (1 + \rho_L) (1 + \rho_S \rho_L + \rho_S^2 \rho_L^2 + \rho_S^3 \rho_L^3 + \rho_S^4 \rho_L^4 + \rho_S^5 \rho_L^5 + \dots)$$

The second bracket is a simple geometric sum, and since $|\rho_L| < 1$ and $|\rho_S| < 1$ this reduces to

$$V_L = \frac{V_0^+ (1 + \rho_L)}{1 - \rho_S \rho_L} = V_S \frac{Z_L}{Z_S + Z_L} \quad (\text{after some simple, but tedious, algebra})$$

This is the expected load voltage in the steady state. The limiting value of load current can be found in a similar way (left as an exercise, but beware of the sign changes when calculating the current).

Problems 9 – Antennas

* = confidence builder, ** = exam standard, *** = harder questions, or extracurricular

Note 1: outline numerical solutions are given on the last page of each problem sheet.

Note 2: all numerical answers should be given to 3 s.f., based on the precision of values in questions.

Note 3: four years of past exam questions are provided in a separate document; “Antennas” appears as part of Q3 on these exam papers (the other part of Q3 being on “EM Waves”; I do not issue answers to exam questions but will cover their solution in the revision lectures leading up to the exam.

AP, April 2024

9.1* (See worked example 9.1) Calculate the radius of the far-field zone of the following antennas: (a) a 9 GHz X-band rectangular horn antenna, of aperture size 138mm by 107 mm; (b) a 4×3 array of parallel 1.8 GHz half-wave dipoles, each spaced a distance of $0.4\lambda_0$ apart.

9.2* (See Section 9.4) - an exercise on dB, dBW and dBm units and conversions

- (a) Convert the following power ratios into dBs: 838, 0.0375, 7.24×10^{-9} .
- (b) Convert the following power ratios into linear units: +63.9dB, -33.5dB, -0.236dB.
- (c) Convert the following powers into both dBm and dBW: 55.2W, 4.05nW, 73.4mW.
- (d) Convert the following powers into W (i.e. the S.I. unit): +8.15dBW, -41.8dBW, -27.4dBm

9.3* (See worked example 9.2) The rms power flux a distance R into the far-field from an isotropic 2.4GHz Tx antenna is $65.8 \mu\text{W}/\text{m}^2$. If its total rms radiated power is 30.5W, calculate R and the EM field magnitudes at this point. Repeat these calculations a further 350m away.

9.4 (See worked example 9.3)** A 5.6 GHz antenna radiates an rms power of 40 W. Assuming it to be an isotropic antenna, calculate the rms power flux 25 m away from it, together with the E and H field magnitudes. Also write down the phasor form of these components at some general distance r from the antenna (assuming this to be in the far-field zone).

9.5 (See worked example 9.4)** The antenna of problem 9.4 actually has a directivity of 6 dBi and a radiation efficiency of 0.935. How much power is dissipated within it and how much electrical power is delivered to it? What is its gain? Also calculate the power flux along its main lobe, 25 m away from the antenna, and the associated field magnitudes. Re-calculate the power flux and field magnitudes 250 m from it, along the same direction.

9.6** The electric field magnitude 90 km from an antenna in the direction of its main beam is found to be 235 $\mu\text{V}/\text{m}$. If the antenna’s directivity is 4.2 dBi, what is its rms radiated power?

9.7 (See worked example 9.5)** A 12GHz transmitter comprises a +9.0dBm source, which is impedance matched to a +42.2dB power amplifier, itself matched to an antenna of 32.5dBi gain. The cables and connectors have 1.6dB attenuation. Calculate the power flux of the antenna’s main beam at a distance of 155km. Heavy snow near the transmitter introduces a further 5.7dB attenuation into the path loss. By what factor is the power flux at 155km reduced?

9.8* (See worked example 9.6) A half wavelength dipole antenna radiates 5 W rms at 2.5 GHz. How long is this antenna? If its directivity is 1.64, calculate its average radiation intensity (i.e. if it were considered to be isotropic) and its maximum radiation intensity. Sketch its radiation pattern, indicating the direction of the main beam relative to the dipole’s axis.

9.9 (See worked example 9.6)** A resonant monopole antenna is 2.7m long and is oriented vertically above some flat ground. Calculate its operating frequency and sketch its radiation pattern. Show by simple consideration of the EM fields generated by a monopole antenna that its directivity must be twice that of an equivalent dipole antenna without the ground plane (i.e. a 3 dB increase in gain).

9.10 (See worked example 9.7)** Estimate the maximum directivity and main beamwidth (in degrees) of the two antennas in problem 9.1. Show the directions of their main beams.

9.11* Estimate the effective area for an Rx antenna made from the dipole of problem 9.8, and the monopole of problem 9.9.

9.12 (See worked example 9.8)** A receiver has a directivity of 23.5dBi, radiation efficiency 0.915 and is exposed to 5.8 GHz microwaves with a uniform rms power flux of 33.5 nW/m^2 , directed along its main beam. Calculate the rms detected power if output matched. The transmitter has a directivity of 16.4 dBi, radiation efficiency 0.88, and is a distance of 8.2 km away from the receiver. How much rms power is being provided to the (matched) Tx antenna and how much rms power is dissipated in it?

9.13 (See worked examples 9.8 and 9.9)** Identical 1.75 GHz antennas (of directivity 18 dBi and radiation efficiency 0.92) are aligned in the direction of their main lobes and are placed on the top of two towers, a distance of 15 km apart. If 1.3 kW rms power is fed into the transmit antenna, calculate the radiated power, the power flux at the receive antenna and estimate the received signal.

9.14 (See worked example 9.10)** A 5.8GHz wireless uplink has a +5.0dBm source amplified by a +44.2dB, which is impedance matched to a Tx antenna of 37.5dBi gain. The uplink's cables and connectors have a total attenuation of 3.0dB. The Rx antenna is 110km from the source and has a gain of 28.5dBi. The Rx antenna cables and connectors have 1.4dB attenuation. Calculate the power transferred to the Rx electronics.

9.15 (See worked example 9.11)** In problem 9.14, if the minimum detectable signal of the Rx electronics is -63.2 dBm , estimate the maximum working range of this wireless link. If better Tx and Rx cables are used to reduce the total attenuation to 2.6dB, how does this affect the maximum range?

9.16 (See Section 9.15)** A 28 GHz link connects a base station with its switching office 3 km away, with transmit and receive antennas both identical, each of 35 dBi gain. Estimate the ratio of received and transmit powers, expressing your answer in dB. Compare this radio link with a wired (coaxial) link, of attenuation 0.05 dB/m with three 20 dB repeaters along its length.

9.17** There is only one examinable derivation in this section, namely the Friis equation for a wireless communications link. Make sure you can derive it based on the information given in the formula sheet, following the steps in Section 9.14.

9.18*** Here are some final questions for discussion and for developing a deeper understanding.

- Consult any advanced text on EM theory to note how the near and far EM fields of an antenna are derived from the wave equation. Also note how the radius of the far field zone is derived.
- What are the problems (if any) of placing an Rx antenna in the near field of a Tx antenna, e.g. as in a Bluetooth connection between nearby portable devices?
- Show that a satellite will be in a geostationary orbit when around 37000 km above the surface of the earth. What is the orbital period of the ISS?
- The typical angular separation of geostationary satellites is about 0.1° . What is the minimum size of earth-bound dish needed to set up a unique 12 GHz link with one of these satellites?
- A parabolic reflector will produce a highly directional main beam. How is the beamwidth of such an antenna linked to diffraction theory?
- Radio waves can be bounced off the moon and detected back on earth using the same high-gain Tx-Rx antenna. Estimate the Rx power of such a “moon-bounce” for a given Tx power and frequency, for a parabolic reflector of a given size.
- When an object is heated it first glows red, then orange, yellow and white (an example being the sun, at a surface temperature of 5800K); why doesn't it go from yellow to green then blue?

Note: *There are no numerical problems on thermal radiation as this section is non-exam for EN2076. However, this section should be of relevance to other modules you may be taking so you are advised to take on board the relevant concepts.*

Problems 9 – Outline Answers to Numerical Problems

9.1(a) $\lambda_0 = c/f = 3.33$ cm, largest antenna dimension $L = 0.138$ m, so R_{ff} is the larger of $2\lambda_0 = 6.67$ cm or $2L^2/\lambda_0 = 1.14$ m, hence $R_{ff} \approx 1.14$ m; (b) $\lambda_0 = c/f = 16.7$ cm, largest antenna dimension $L = 3 \times 0.4\lambda_0 = 0.2$ m, so R_{ff} is the larger of $2\lambda_0 = 33.3$ cm or $2L^2/\lambda_0 = 48$ cm, hence $R_{ff} \approx 0.5$ m.

9.2(a) 29.2dB, -14.3dB, -81.4dB; (b) 2.45×10^6 , 4.47×10^{-4} , 0.947; (c) 17.4dBW or 47.4dBm, -83.9dBW or -53.9dBm, -11.3dBW or 18.7dBm; (d) 6.53W, 6.61×10^{-5} W, 1.82×10^{-6} W.

9.3 Since $S = P_{rad}/4\pi R^2 \rightarrow R = \sqrt{P_{rad}/4\pi S} = 192$ m, field magnitudes are $E_0 = \sqrt{2\eta_0 S} = 0.223$ V/m, $H_0 = E_0/\eta_0 = 5.91 \times 10^{-4}$ A/m; if R is increased from 192m to 542m, S reduces by a factor $(542/192)^2$ to $8.26 \mu\text{W/m}^2$, the field magnitudes both reduce by a factor $(542/192)$ to $E_0 = 79.0$ mV/m and $H_0 = 2.09 \times 10^{-4}$ A/m.

9.4 This is an isotropic antenna, $S = P_{rad}/4\pi r^2 = E_0^2/2\eta_0 \rightarrow E_0 = \sqrt{P_{rad}\eta_0/2\pi r^2} = 1.96$ V/m, $H_0 = E_0/\eta_0 = 5.20 \times 10^{-3}$ A/m. The $\underline{E}$ and $\underline{H}$ magnitudes both have the phasor forms $(A/r)\exp j(\omega t - \beta r)$ with $\underline{E}$ perpendicular to $\underline{H}$ and perpendicular to $\hat{a}_r$, $\omega = 3.52 \times 10^{10}$ s $^{-1}$ and $\beta = 117$ m $^{-1}$. The value of the constant A can be deduced from $A = E_0 \times 25\text{m} = 49.0$ V.

9.5 $P_{rad} = e P_{in} \rightarrow P_{in} = 42.8$ W, rms power dissipated in the Tx antenna is $42.8 \text{ W} - 40 \text{ W} = 2.8 \text{ W}$; $D = 6\text{dBi} \equiv 10^{6/10} = 3.98$ (linear units), so $G = e D = 3.72$ or $10\log(3.72) = 5.71$ dBi; at a distance of 25 m the rms power flux is $S = D P_{rad}/4\pi r^2 = 20.3 \text{ mW/m}^2$ so $E_0 = \sqrt{2\eta_0 S} = 3.91$ V/m and $H_0 = E_0/\eta_0 = 1.04 \times 10^{-2}$ A/m; at a distance of 250m, rms power flux is 0.203 mW/m^2 (i.e. 100 times less), $E_0 = 0.391$ V/m, $H_0 = 1.04 \times 10^{-3}$ A/m (both field magnitudes are 10 times less).

9.6 $D = 4.2\text{dBi} \equiv 10^{4.2/10} = 2.63$; since $S = D/4\pi r^2 = E_0^2/2\eta_0 \rightarrow P_{rad} = 2\pi r^2 E_0^2/D\eta_0 = 2.83$ W.

9.7 The rms input power to the Tx antenna is $P_{in} = 9.0 + 42.2 - 1.6 = 49.6\text{dBm} = 19.6\text{dBW} = 91.2\text{W}$. The Tx antenna gain is $32.5\text{dBi} = 1778$ and at $r = 155\text{km}$ the main beam power flux is $S = G_t P_{in}/4\pi r^2 = 5.37 \times 10^{-7} \text{ W/m}^2$; 5.7dB attenuation reduces S by the factor $10^{-0.57} = 0.269$.

9.8 Since $\lambda_0 = c/f = 12$ cm, antenna length is $\ell = \lambda_0/2 = 6$ cm; the average radiation intensity is $U_{av} = P_{rad}/4\pi = 0.398$ W, the maximum radiation intensity is simply D times this, i.e. $U_{max} = D P_{rad}/4\pi = 0.653$ W. See Section 9.9 for plots of its radiation pattern.

9.9 The monopole length is $\ell = \lambda_0/4$, so $\lambda_0 = 10.8\text{m} \rightarrow f = c/\lambda_0 = 27.8\text{MHz}$. All the radiated power is now confined to the space above the ground plane, thus doubling the directivity (and gain); this factor of 2 is equivalent to $10\log 2 = 3.01$ dBi. The monopole's directivity is therefore $1.76 \times 2 = 3.52$, or $2.15\text{dBi} + 3.01\text{dB} = 5.16$ dBi.

9.10 (a) $\lambda_0 = c/f = 3.33\text{cm}$, $A = 0.0148\text{m}^2 \rightarrow D_{max} \approx 4\pi A/\lambda_0^2 \approx 167 \equiv 22.2\text{dBi}$; beam angle

$\Delta\theta \approx \Delta\phi \approx \sqrt{4\pi / D_{\max}} \approx 0.27 \text{ rads} \equiv 16^\circ$ (b) $D_{\max} \approx 4\pi A / \lambda_0^2 \approx 4\pi(3 \times 0.4 + 2 \times 0.4) \approx 25 \equiv 14 \text{ dBi}$;
beam angle $\Delta\theta \approx \Delta\phi \approx \sqrt{4\pi / D_{\max}} \approx 0.71 \text{ rads} \equiv 40^\circ$. The main beams are perpendicular to their areas.

9.11 These two antennas have no “aperture” as such, but still present an effective area for power collection to an incoming EM wave. The effective area is $A_{\text{eff}} \approx D\lambda_0^2 / 4\pi$. For the half-wave dipole antenna of problem 9.6, $\lambda_0 = 12 \text{ cm}$ and $D = 1.76$, giving $A_{\text{eff}} \approx D\lambda_0^2 / 4\pi \approx 0.0081 \text{ m}^2$ (81 cm^2); for the resonant monopole of problem 9.7, $\lambda_0 = 10.8 \text{ m}$ and $D = 3.52$, giving $A_{\text{eff}} \approx D\lambda_0^2 / 4\pi \approx 33 \text{ m}^2$.

9.12 The directivity of the Rx antenna $D_r = 10^{2.35} = 224$ and the free space wavelength is $\lambda_0 = 5.17 \text{ cm}$, so the effective area of the Rx antenna is $A_{\text{eff}} \approx D\lambda_0^2 / 4\pi \approx 0.0476 \text{ m}^2$. The Rx power received is $P_{\text{rec}} = S_t A_{\text{eff}} = 1.60 \times 10^{-6} \text{ W}$ and the power output that the Rx antenna delivers to its (matched) detector is $P_{\text{out}} = e_r P_{\text{rec}} = 1.46 \times 10^{-9} \text{ W}$. The directivity of the Tx antenna is $D_t = 10^{1.64} = 43.7$ so the Tx gain is $G_r = e_t D_t = 38.4$. The Tx power flux at a distance r is $S = G_t P_{\text{in}} / 4\pi r^2 \rightarrow P_{\text{in}} = 4\pi r^2 S / G_t = 0.737 \text{ W}$. The radiated power is $P_{\text{rad}} = e_t P_{\text{in}} = 0.649 \text{ W}$, so the dissipated power in the Tx antenna is $P_{\text{diss}} = P_{\text{in}} - P_{\text{rad}} = 0.737 \text{ W} - 0.649 \text{ W} = 88 \text{ mW}$.

9.13 Tx radiated power is $P_{\text{rad}} = e_t P_{\text{in}} = 1.20 \text{ kW rms}$, $D_t = D_r = 18 \text{ dBi} \equiv 10^{18/10} = 63.1$; power flux at receiver $S = D_t P_{\text{rad}} / 4\pi r^2 = 2.68 \times 10^{-5} \text{ W/m}^2$; the effective area of the Rx antenna is $A_{\text{eff}} = D_r \lambda_0^2 / 4\pi = 0.147 \text{ m}^2$ (since $\lambda_0 = 0.171 \text{ m}$), so the power at the Rx antenna is approximately $P_{\text{rec}} \approx A_{\text{eff}} S \approx 3.94 \text{ } \mu\text{W}$ and 92% of this is delivered to the Rx detector (if matched), i.e. $P_{\text{out}} = e_r P_{\text{rec}} = 3.62 \text{ } \mu\text{W}$

9.14 $\lambda_0 = c / f = 5.26 \text{ cm}$ so the free space path loss is $-20 \log(\lambda_0 / 4\pi r) = -148.4 \text{ dB}$. The link budget for the rms Rx power (in dBm) is $+5.0 + 44.2 - 3.0 + 37.5 - 148.4 + 28.5 - 1.4 = -37.6 \text{ dBm} \equiv 174 \text{ nW}$.

9.15 Treat the free space path loss PL as a variable and set the range to be at its maximum value, in which case the link budget for Rx power is $+5.0 + 44.2 - 3.0 + 37.5 + \text{PL} + 28.5 - 1.4 = -63.2$. Hence, the minimum path loss allowed is -174.0 dB , so the maximum range R is determined calculated from $-20 \log(\lambda_0 / 4\pi R) = -174.0 \rightarrow R \approx 2098 \text{ km}$ (subject to the approximations listed in Section 9.14). If better cables reduce the total attenuation from 4.4 dB down to 2.6 dB , that means a 1.8 dB increase in the Rx power, so the minimum path loss for the maximum range is now -175.8 dB . Hence, the maximum range is increased to $R \approx 2581 \text{ km}$ (or by the factor $10^{1.8/20} = 1.23$).

9.16 Using the Friis equation, power ratio (dB) $= G_t + G_r - 20 \log(4\pi r / \lambda_0) = -60.9 \text{ dB}$ (since $\lambda_0 = 1.07 \text{ cm}$); power ratio for coaxial cable without repeaters is $\alpha_r = -150 \text{ dB}$, with repeaters is $(-150 + 20 + 20 + 20) \text{ dB} = -90 \text{ dB}$; hence, the wireless link is much more efficient than the wired link, even with the repeaters (by 29.1 dB or a factor of 813).